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EXPLORING COLD POOL DYNAMICS IN MONAN: INSIGHTS FROM HURRICANES

Bianca Fusinato

Master's Dissertation from the
Graduate Program in Meteorology,
supervised by Dr. Saulo R. Freitas,
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**ATENÇÃO! A FOLHA DE
APROVAÇÃO SERÁ INCLU-
IDA POSTERIORMENTE.**

Mestrado ou Doutorado em Nome do
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To my beloved family and dear friends

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ABSTRACT

A new cold pool parameterization, coupled with a traditional mass flux convection parameterization (Grell-Freitas), was evaluated in the upcoming next generation of the Brazilian weather and climate forecast system called Model for Ocean-laNd-Atmosphere predictionN (MONAN). The new system is based on the Model for Prediction Across Scales (MPAS). This work aims to assess the impact of this parameterization on the trajectory, intensity, and rainfall associated with two tropical cyclones from the 2024 hurricane season, Beryl and Helene. For Hurricane Beryl, a set of sensitivity experiments was conducted to assess the influence of initial conditions, cold pool lifetime, vertical displacement of maximum mass flux, and model resolution (15, 30, and 60 km). The results demonstrated that the inclusion of cold pool dynamics significantly improved trajectory accuracy, particularly at later forecast hours, and enhanced rainfall representation, especially regarding the spatial structure of spiral rainbands and extreme precipitation. Wind intensity forecasts also benefited from a more realistic convective structure, despite persistent pressure biases. An optimal configuration combining reduced cold pool lifetime (1 hour), displaced mass flux peak height, and higher resolution yielded balanced performance across all metrics. This configuration was then applied to Hurricane Helene, where it successfully captured the observed extreme rainfall over the southeastern United States, with improved spatial distribution and overall intensity, and trajectory accuracy. These findings suggest that key cold pool parameters have a substantial impact on tropical cyclone simulations within MONAN, particularly for rainfall forecasting.

Keywords: Cold Pool. Parameterization. MONAN. Hurricanes. Beryl. Helene.

EXPLORANDO A DINÂMICA DE *COLD POOLS* NO MONAN: PERSPECTIVAS A PARTIR DE FURACÕES

RESUMO

Uma nova parametrização de *cold pool*, acoplada a uma parametrização convectiva tradicional de fluxo de massa (Grell-Freitas), foi avaliada na próxima geração do sistema brasileiro de previsão do tempo e clima, denominado *Model for Ocean-laNd-Atmosphere predictioN* (MONAN). O novo sistema é baseado no *Model for Prediction Across Scales* (MPAS). Este trabalho tem como objetivo avaliar o impacto dessa parametrização na trajetória, intensidade e chuva, associadas a dois ciclones tropicais da temporada de furacões de 2024: Beryl e Helene.

Para o furacão Beryl, foi realizado um conjunto de experimentos de sensibilidade para avaliar a influência das condições iniciais, tempo de vida das *cold pools*, deslocamento vertical do pico de fluxo de massa e resolução do modelo (15, 30 e 60 km). Os resultados demonstraram que a inclusão da dinâmica da *cold pool* melhorou significativamente a precisão da trajetória - especialmente nas horas mais avançadas da previsão - e aprimorou a representação da chuva, com destaque para a estrutura espacial das bandas de chuva em espiral e eventos extremos de chuva. As previsões de intensidade dos ventos também se beneficiaram de uma estrutura convectiva mais realista, apesar de persistirem vieses de pressão.

Uma configuração ótima, combinando tempo de vida reduzido do *cold pool* (1 hora), deslocamento da altura do pico de fluxo de massa e maior resolução, apresentou desempenho equilibrado em todas as métricas. Essa configuração foi então aplicada ao furacão Helene, onde conseguiu representar com sucesso a chuva extrema observada no sudeste dos Estados Unidos, com melhor distribuição espacial e maior acurácia nas previsões de intensidade e trajetória.

Esses resultados indicam que parâmetros-chave relacionados à *cold pool* têm impacto substancial nas simulações de ciclones tropicais no MONAN, particularmente na previsão de chuva.

Palavras-chave: Cold Pool. Parametrização. MONAN. Furacões. Beryl. Helene.

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LIST OF ABBREVIATIONS

ACE	– Accumulated Cyclone Energy
ALE	– Available Lifting Energy
ALP	– Available Lifting Power
API	– Application Program Interface
ATE	– Along-Track Error
BRAMS	– Brazilian development on the Regional Atmospheric Modeling System
CAPE	– Convective Available Potential Energy
CDF	– Cumulative Density Function
CDO	– Climate Data Operators
CDS	– Cloude Data Store
CIN	– Convective Inhibition Energy
CMORPH-KF	– CPC Morphing-Kalmann Filter
CONUS	– Continental United States
CP	– Cold Pool
CPP	– Cold Pool Perturbation
CPC	– Climate Prediction Center
CRM	– Cloud-Resolving Model
CTE	– Cross-Track Error
DISC	– Data and Information Services Center
DPR	– Dual-frequency Precipitation Radar
ECMWF	– European Centre for Medium-range Weather Forecasts
EMC	– Environmental Modeling Center
ERA5	– fifth generation of atmospheric reanalysis conducted by the European Centre for Medium-Range Weather Forecasts
ETS	– Equitable Threat Score
GCMs	– Global Circulation Models
GES	– NASA Goddard Earth Science
GF	– Grell-Freitas
GF-CP	– Grell-Freitas-Cold Pool
GFS	– Global Forecast System
GISS	– Goddard Institute for Space Studies
GMS	– Geosynchronous Meteorological Satellite
GOES	– Geosynchronous Operational Environmental Satellites
GPM	– Global Precipitation Measurement
GPM-MERGIR	– NASA GES DISC GPM Merged 4-Km IR Tb dataset
GSMaP	– Global Satellite Mapping of Precipitation
HAFS	– Hurricane Analysis and Forecast System
IBTrACS	– International Best Track Archive for Climate Stewardship
IC	– Initial Conditions
IFS	– ECMWF Integrated Forecast System
IMERG	– Integrated Multi-satellite Retriavals for GPM
INPE	– Brazilian National Institute for Space Research

IR	– Infrared
ISU	– Yonsei University
LCL	– Lifting Condensation Level
LES	– Large Eddy Simulation
LFC	– Level of Free Convection
MAE	– Mean Absolute Error
MCTI	– Ministry of Science, Technology and Innovation
Meteosat	– Meteorologica Satellite
MJO	– Madden-Julian Oscillation
MONAN	– Model for Ocean-laNd-Atmosphere prediction
MPAS	– Model for Prediction Across Scales
MSLP	– Mean Sea Level Pressure
MSE	– Moist Static Energy
MTSat	– Multi-functional Transport Satellite
NCAR	– National Center for Atmospheric Research
NCEP	– National Centers for Environmental Prediction
NWS	– National Weather Service
PBL	– Planetary Boundary Layer
PDIR	– PERSIAMN Dynamic Infrared-rain Rate
PERSIAMN	– Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks
PMM	– Precipitation Measuring Mission
PDF	– Probability Density Function
RFCs	– River Forecast Centers
RMSA	– Root Mean Square Error
SHOC-MF	– Simplified High Order Closure-Mass Flux
Tb	– Brightness Temperature
TCs	– Tropical Cyclones
TKE	– Turbulent Kinetic Energy
UK	– United Kingdom
UM	– Met Office Unified Model
US	– United States
WPAS	– WRF Pre-processing System
WRF	– Weather Research Forecasting Model

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1 INTRODUCTION

Global Circulation Models (GCMs) play a fundamental role in climate and weather forecasting, as they are capable of simulating the complex physical interactions between atmospheric, oceanic, and terrestrial systems. However, due to their global coverage, running these models at high spatial resolution requires substantial computational power. As a result, only a limited number of research centers worldwide are capable of producing these high-resolution simulations. Even in such cases, many key atmospheric processes occur at spatial and temporal scales smaller than the model grid resolution and thus cannot be explicitly resolved. These are referred to as subgrid-scale processes, and their representation relies on parameterization schemes (STENSRUD, 2007), (ARAKAWA et al., 2011), (ARAKAWA; SCHUBERT, 1974), (ARAKAWA, 2004).

Parameterizations are mathematical frameworks designed to approximate the effects of unresolved processes within the model grid. Among them, convection schemes are essential, as they aim to represent the net effect of convective clouds and associated phenomena such as updrafts, downdrafts, detrainment, and cold pools. In particular, downdrafts, the descending currents of cold, dense air, are commonly included as part of convective parameterizations (GRELL; FREITAS, 2014). Both downdrafts and the precipitation evaporation give rise to cold pools, which are the focus of this dissertation.

Cold pools contain cool and dry air that descends from convective downdrafts and spreads horizontally upon reaching the surface. As they propagate, they modify the surrounding environment both thermodynamically and mechanically. Specifically, cold pools displace the warmer, moister air in the boundary layer and create low-level convergence at their leading edge, thereby enhancing the likelihood of new convective cell formation (TOMPKINS, 2001), (TORRI et al., 2015), (FENG et al., 2015), (FREITAS et al., 2024). This mechanism contributes not only to the maintenance and propagation of mesoscale convective systems, such as squall lines, but also to the organization of deep convection in tropical environments (FENG et al., 2015), (FREITAS et al., 2024).

The present work focuses on the evaluation of a cold pool parameterization coupled within the Grell-Freitas convection scheme. One of the key components of this scheme is the buoyancy-excess coefficient, which is used to diagnose the presence of cold pools in the model. This coefficient relates moist static energy anomalies inside the downdraft and the environment. When conditions indicate the existence of a

cold pool, the parameterization simulates the associated gust front by modifying the horizontal wind and potentially triggering new convective events depending on the available vertical kinetic energy (FREITAS et al., 2024).

Given their role in initiating and organizing convection, cold pools may also influence the development and internal structure of tropical cyclones. These extreme weather systems are responsible for substantial socioeconomic impacts, particularly due to heavy rainfall, storm surges, and destructive winds. During the 2024 hurricane season in the United States (US), the financial repercussions of hurricane-related damages were estimated to be approximately \$24 billion (National Oceanic and Atmospheric Administration (NOAA), 2025b; National Oceanic and Atmospheric Administration (NOAA), 2025a). This substantial economic impact underscores the significant vulnerabilities associated with extreme weather events and their implications for both regional and national economies. Therefore, improving the simulation of such systems is a high priority in both weather and climate prediction efforts.

Tropical cyclones are defined as organized, rotating low-pressure systems that form over warm ocean waters. They are called hurricanes in the North Atlantic, typhoons in the North Pacific, and cyclones in the South Pacific and Indian Ocean. Although much attention has been devoted to the role of large-scale environmental factors in cyclone formation and intensification, recent studies have highlighted the importance of mesoscale processes, including cold pools, in modulating rainband formation and the evolution of convective structures within these storms (SAWADA; IWASAKI, 2010a; CHEN et al., 2014; GRANT; HEEVER, 2018; CAI; TANG, 2019).

To explore how cold pool parameterizations influence tropical cyclone simulations, this dissertation adopts the Model for Ocean-land-Atmosphere predictionN (MONAN) system, which is currently under development by the Brazilian National Institute for Space Research (INPE). MONAN belongs to a new generation of atmospheric models based on the Model for Prediction Across Scales (MPAS) dynamic core, which uses an unstructured mesh. This mesh configuration enables seamless simulations across nested or variable-resolution domains, reducing numerical artifacts commonly associated with boundary conditions in traditional regional models (SKAMAROCK et al., 2012).

The primary objective of this study is to evaluate the cold pool parameterization in MONAN during the simulation of tropical cyclones, with a focus on its impact on trajectory, intensity, and rainfall fields. Moreover, we aim to assess whether adjustments to key parameters, such as the cold pool lifetime, can lead to improvements

in the model’s ability to forecast tropical cyclone intensity and structure. Through this effort, the dissertation seeks to contribute to the ongoing development of convection parameterizations and enhance the skill of high-impact weather forecasting in tropical regions.

1.1 General objective

To explore a cold pool parameterization coupled with the Grell-Freitas (GF) convection parameterization within the Model for Ocean-laNd-Atmosphere prediction (MONAN).

1.2 Specific objectives

- a) Assess the impact of the cold pool parameterization scheme on the forecasted trajectory, intensity, and rainfall of hurricanes.
- b) Perform sensitivity experiments within a hurricane case study (namely, Hurricane Beryl - 2024), investigating the influence of initial conditions, cold pool duration, the displacement of the maximum mass flux height, and resolution effects.
- c) Identify the optimal parameter configuration for the cold pool parameterization to be used in MONAN.
- d) Evaluate the performance of this optimal configuration in Hurricane Helene (2024), alongside resolution effects..

1.3 Scientific questions

This work builds upon the findings and open issues identified in [Freitas et al. \(2024\)](#). The following scientific questions are proposed:

- Does the GF-Cold Pool (GF-CP) scheme improve the forecast of hurricane trajectory, intensity, and rainfall in MONAN?
- Which parameters within the GF-CP have the most significant influence on hurricane representation in MONAN?
- How robust is the optimal GF-CP configuration when applied to a different hurricane event?
- Can improvements in cold pool representation lead to better rainfall predictions in tropical cyclones?

1.4 Thesis organization

This dissertation is organized as follows: Chapter 2 presents a literature review on the subject of this work, beginning with the concept of convective clouds and the associated cold pool phenomenon, followed by the parameterization of this phenomenon, and concluding with an overview of hurricane basic concepts and the role of cold pools within. The data used and the model that performed the forecasts are described in Chapter 3. Chapter 4 outlines the methodology employed, which is primarily based on analyzing hurricane forecasts through three categories: track, intensity, and rainfall. This chapter concludes with a workflow diagram that is applied to each of the case studies. Chapters 5 and 6 present the results and analyses of the two case studies. Finally, Chapter 7 contains the conclusions and potential projects that may be pursued following this dissertation.

2 LITERATURE REVIEW

This chapter begins with an overview of the conceptual framework of an air parcel that evolves into a convective cloud, highlighting the mechanical and thermodynamic processes involved in this transformation (Section 2.1). One of the outcomes of convection is the formation of cold pools, which are discussed in detail in Section 2.2. Since this dissertation is intrinsically linked to atmospheric modeling, in particular the modeling of convective processes, a description of the cold pool parameterization is provided in Section 2.3, along with a brief explanation of the Grell-Freitas parameterization coupling, addressed in Section 2.4. Given that one of the objectives of this study is to assess the impact of this parameterization in hurricane case studies, Section 2.5 provides an overview of the main characteristics of the hurricanes analyzed, as well as key aspects of cold pools within. Finally, Section 2.6 summarizes the main points discussed in this chapter, which serve as a foundation for the analyses presented in the following chapters.

2.1 Overview of convective cloud dynamics

In convective cloud dynamics, various complex analyses can be employed, yet some may potentially reveal theoretical gaps. To address this situation, [Muller and Takayabu \(2020\)](#) offer a simplified view of the stages in the convective cloud life cycle, which can be condensed in the following paragraphs. It is essential to understand that in convective clouds, convection is the prevailing process shaping the cloud's state. We refer to a convective cloud as a type of cloud that exhibits vertical development, such as cumulonimbus clouds.

Convection is the transport of sensible and latent heat from hotter and more humid areas to cooler and drier areas, being one of the three primary methods of heat transfer, alongside radiation and conduction. The focus, when meteorologists refer to convection, is on the fluid movement and transport associated with buoyancy.

Buoyancy is defined as a force created by a temperature (or virtual temperature) difference. For example, inside a pan containing boiling water, the water next to the bottom is hotter than the upper water layers. Heated water below starts to rise, popping up on the top (boiling). The buoyant force causes the warmer water below to rise and float upward because of the temperature differences. When rising, the water carries the energy accumulated at the bottom, leading it to the upper layers, which means it is occurring convection inside the pan. One can state that buoyancy defines convection because it must contain a temperature gradient that will lead the

heat to low-energy areas.

The example of boiling water in a pan is a simplified representation of the atmosphere. In the atmosphere, an air parcel rises due to sunlight heating the Earth's surface. The surface will be hotter than the atmosphere above, leading to a temperature gradient, and then, the air near the surface will start to rise (float), transporting that energy. When this happens, one can say that the floating air has positive buoyancy.

Summarizing, buoyancy is the property that induces the transport mechanism responsible for carrying heat, moisture, and momentum from the surface to the atmosphere and is one of the main agents for cloud formation.

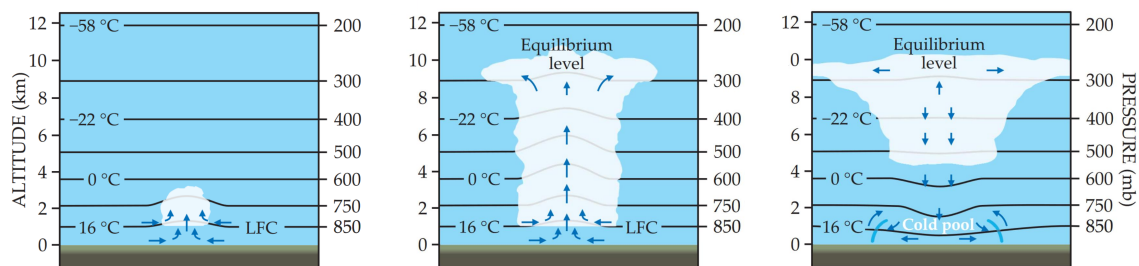
The upward movement also carries near-surface water vapor and many other tracers in the form of gases and particulates. The vertical air current formed is called an updraft and is an important mechanism that maintains the clouds. Then, adiabatic expansion (expansion ideally without energy loss) causes the rising moist air to cool as the pressure drops at higher altitudes. At a certain height, temperature and pressure continuously decrease, allowing the water vapor to condense (become saturated), depending on temperature. That level in the atmosphere is called the Lifting Condensation Level (LCL). Those tiny suspended condensates form a visible cloud.

When a moistened air parcel holds sufficient buoyancy to exceed the negatively buoyant barrier imposed by its environment, it proceeds to ascend until it reaches a state of free buoyancy, known as the Level of Free Convection (LFC). The buoyant barrier is defined as the Convective Inhibition Energy (CIN), which is the energy that is required to be given to the air parcel for its initial movement from the surface to the LCL. So, if the CIN is high, the air parcel will not reach the LCL and thus cannot form a cloud.

Above the LFC, the air parcel continues to rise because it no longer has the CIN to obstruct the rise. If the air parcel is rising, then it has positive buoyancy. A practical measure of the total possible buoyant energy in an environment is the Convective Available Potential Energy (CAPE). The bigger the CAPE values, means bigger the buoyancy is, leading the air parcel to higher vertical levels. Notably, deep convection can occur when an air parcel contains large values of CAPE and either values of CIN are small or the presence of some synoptic systems, which may act as a trigger to release the energy, or both.

Now, the air parcel is free to rise until the inside density matches the environmental density, and the cloudy air is no longer buoyant - this stage is also described as the mature stage of cloud formation. The two leftmost images in Figure 2.1 illustrate this process.

Figure 2.1 - Cloud stages of formation.



Source: Muller and Takayabu (2020).

When a deep convective cloud reaches zero buoyancy in the higher troposphere, it means that the environmental density is equal to the cloud density. Therefore, the equilibrated densities form a barrier-like structure at upper levels of the troposphere, then making the upper-level air spread horizontally, forming what's known as an anvil (see anvil shape at the rightmost image on Figure 2.1). Inside the cloud, microphysical processes take part and describe how the condensed droplets and ice crystals grow. Eventually, when they are large enough, they start to precipitate and fall toward the ground.

The precipitation may partially evaporate (MALLINSON; LASHER-TRAPP, 2019), changing the local thermodynamics, specifically, the amount of latent heat decreases. As some air evaporates, it creates a cold air mass (due to the lack of latent heat), known as a **cold pool** (CP; see the rightmost image on Figure 2.1), that descends in a downward movement and spreads horizontally at the surface. These downward

motions, known as downdrafts, counteract the original updraft motion that started the convective cloud storm, thereby ending the cloud life cycle. Although both play significant roles in cloud dynamics, it has to be recognized that, in consequence of the larger particle size, the evaporation of water in a downdraft can virtually never be complete or as efficient as its condensation in an updraft (KESSLER, 1982).

To understand cold pools, which are the subject matter of this dissertation, some physical quantities are relevant in such a process and were described in this section. Downdrafts and buoyancy are crucial to understanding cold pools' dynamics. When the downdraft reaches the surface, the air mass spreads horizontally at a certain high speed; this phenomenon is called the gust front and can play a significant role in forming new clouds. One could define CP as a negatively buoyant air mass, due to its vertical motion from the upper atmosphere to the surface. Being part of the general deep convection structure, the next section will discuss how those quantities interact to form and organize new convection.

2.2 Cold pools

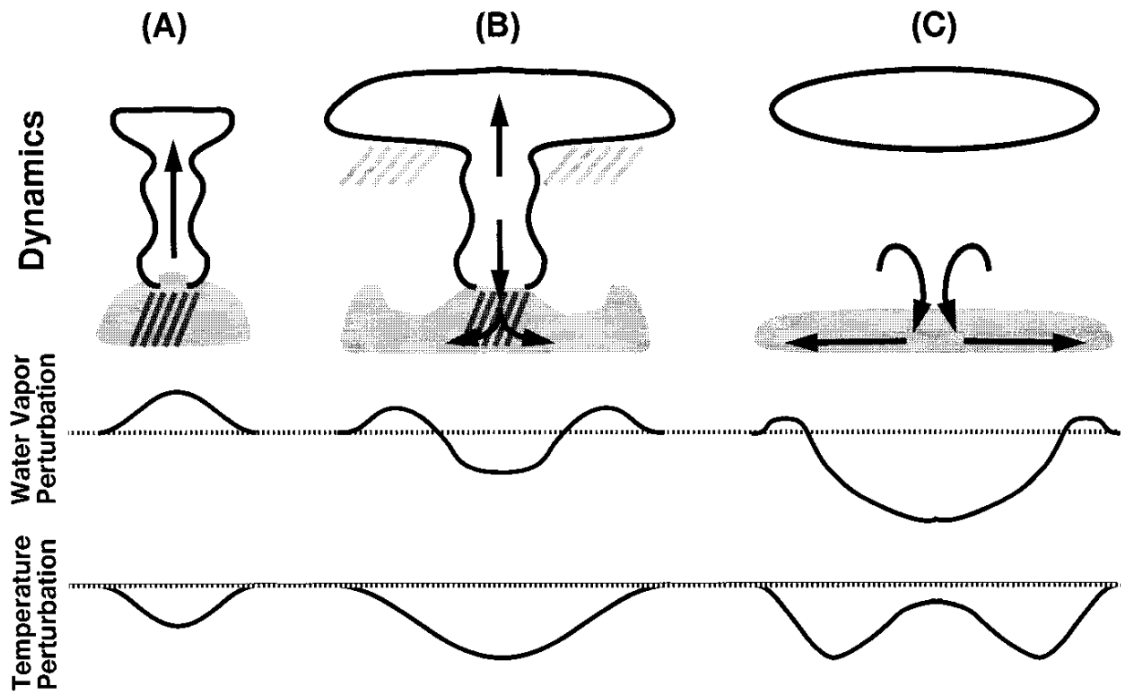
In the previous section, we outlined the location and formation process of a cold pool. As discussed, this phenomenon relies on key elements such as evaporation and significant downdrafts within convective clouds, which can lead to the development of new convective cells through gust front lifting or due to the thermodynamic properties.

The lifting mechanisms described in the previous sentence have been a longstanding issue. Initially, when cold pool studies started, as so-called density currents, the prevailing belief was that the triggering mechanism for initiating new convection involved the linear momentum (mechanical energy) at the edge of the cold pool (WEISMAN; ROTUNNO, 2004; EMANUEL, 1994). It was believed that this process mechanically elevated the air above the boundary layer until it reached the LFC, thus triggering a new cycle of convection. However, according to Tompkins (2001), in scenarios of deep convection over oceans with low vertical wind shear, the mechanism through which convectively induced cold pools initiate new deep convective cells differs significantly from the previously assumed dynamical lifting role. Instead, initiation is attributed to thermodynamic lifting due to anomalies in heat and moisture, leading to a reduction in CIN at the periphery of the cold pool.

In Tompkins (2001), the formation of cold pools is detailed, outlining three distinct phases of its development as analyzed through a Cloud-Resolving Model (CRM) over

tropical oceans. These phases are characterized by conditions of limited large-scale wind shear (low wind shear) and a homogeneous underlying sea surface temperature. In the initial phase, a deep convective tower emerges, causing the subcloud layer beneath the convection to cool and become moistened by precipitation evaporation. This process generates water vapor in the air, which already contains high equivalent potential temperature (θ_e) values. The cooling effect creates a negative buoyancy, resulting from the latent heat consumption during the phase transformation from rain to water vapor, occurring even before the convective downdraft reaches the boundary layer. The temperature (θ_e) is held constant, assuming an adiabatic saturated process. Figure 2.2 (A) shows this initial stage, where water vapor perturbations peak while temperature perturbations remain low.

Figure 2.2 - Stages of cold pool development in a low wind shear environment.



Source: [Tompkins \(2001\)](#).

The second stage begins during the mature phase of a convective cell, when the downdraft (tendency to destroy cloud) propels the negative buoyancy air to the edges of the CP and injects cold and dry air into the center, providing two sources of negative buoyancy to the boundary layer, as shown in Figure 2.2 (B). Simultaneously, the moist band (water vapor) is transported with a negative buoyancy along the

periphery of the recently formed CP, establishing low water vapor in the middle and, conversely, high perturbations around. An intriguing characteristic of the CP structure is the distinct thermal and moisture conditions within its confines; the inner region is drier and colder than the edges, a consequence of the rain evaporation and downdraft’s injection into the boundary layer.

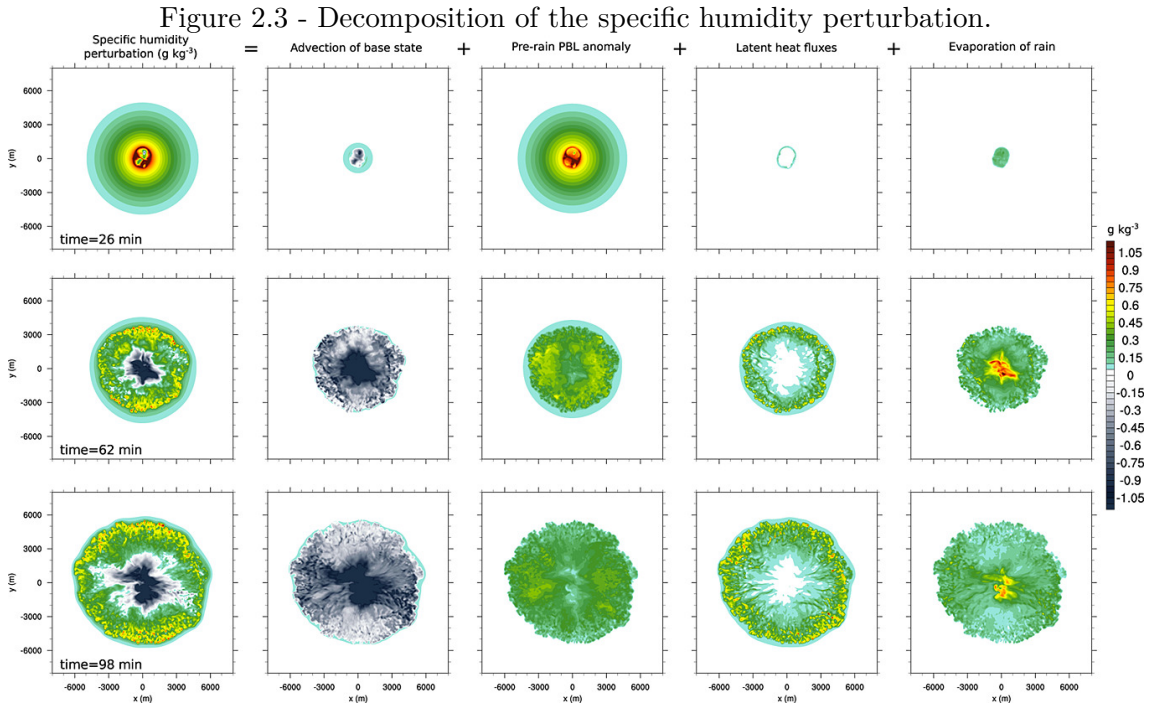
In the dissipation phase of convective cells, cirrus cloud-type persists, as indicated in Figure 2.2 (C). In its final stage on the ground, the CP comes into contact with surface winds, increasing them, and the differences in surface-boundary layer¹ conditions begin to raise both temperature and moisture content, due to the enhancement of surface fluxes promoted by the increasing surface winds. However, entrainment² from the upper boundary layer - upper arrows on Figure 2.2 (C) comes drier and hotter than the downdraft injection, making the moisture content delayed while tending to elevate the temperature at the center of CP. In other words, temperature recovers more quickly than moisture does inside the CP. Alongside this entrainment, sensible and latent heat begin to recover the surface temperature and moisture at the CP’s edges. At the edge, CIN is then reduced as the temperature increases and the moist air tends to lift until it reaches the LCL. Simultaneously, while most of the cold pool regions exhibit considerable CIN, it diminishes to nearly zero as one moves towards the edges. This reduction occurs precisely in the areas where the largest CAPE values are observed. Therefore, the highest CAPE values within the domain are notably linked to the outer extents of the cold pools. Eventually, with low values of CIN, high values of CAPE, and moist air, it will trigger new convection, making the cold pools former’s convective clouds.

Alongside Tompkins (2001), studies by Feng et al. (2015) highlight a colder and drier center, exhibiting significantly lower Moist Static Energy (MSE) than the edge. This distinct feature, consistently observed across CRM studies (LANGHANS; ROMPS, 2015; TORRI et al., 2015; FUGLESTVEDT; HAERTER, 2020), involves a separation of CP properties into the core and the edge. This distinction plays a significant role in understanding trigger mechanisms. The debate on whether the origin of anomalous moisture or MSE is by rain evaporation, surface fluxes, or preexisting moisture anomalies from previous cold pools remains unresolved.

¹“A layer of air of order tens of meters thick adjacent to the ground where mechanical (shear) generation of turbulence exceeds buoyant generation or consumption”. Source: <https://glossary.ametsoc.org/wiki/Surface_boundary_layer>, accessed on 06/19/2025.

²“In meteorology, the mixing of environmental air into a preexisting organized air current so that the environmental air becomes part of the current.” Source: <<https://glossary.ametsoc.org/wiki/Entrainment>>, accessed on 06/19/2025.

For example, in a more recent study by [Langhans and Romps \(2015\)](#), Large Eddy Simulation (LES) techniques were utilized to model the ascent of individual cumulus clouds within a tropical oceanic atmosphere, and the vapor sources were studied using passive and active tracers of water vapor mass. This investigation aimed to seek a physical explanation of the origin of the moisture at the rings of elevated specific humidity observed at the leading edges of CPs. The primary objective was to elucidate the factors contributing to specific humidity perturbations within cold pools. Figure 2.3 illustrates the decomposition of this perturbation.



Source: [Langhans and Romps \(2015\)](#).

According to the authors, surface latent heat fluxes are the primary cause for ring-shaped water vapor anomalies inside the leading edge of a cold pool, while the evaporation of the rain below the cloud base is secondary in their simulations, complementing the study conducted by [Tompkins \(2001\)](#).

As stated by [Tompkins \(2001\)](#), [Feng et al. \(2015\)](#), [Torri et al. \(2015\)](#) argue that lifting is not solely caused by mechanical aspects as previously thought ([WEISMAN; ROTUNNO, 2004](#)); convection can also be thermodynamically triggered by the accumulation of moisture around the edges of cold pools (the moisture rings).

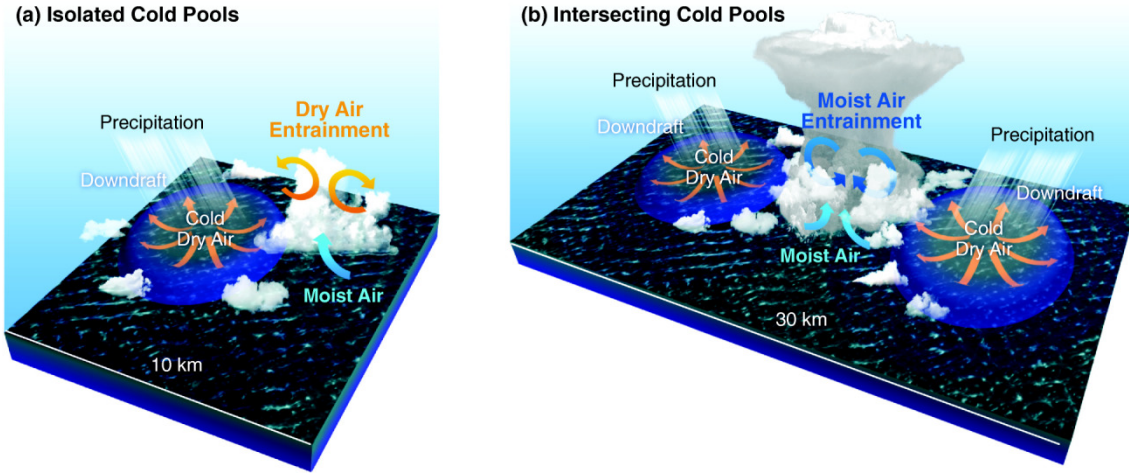
In their study, [Torri et al. \(2015\)](#) proposes a method based on Lagrangian tracking to disentangle the signatures of both forcings (mechanically and thermodynamically) and quantify their significance in a given environment. This study considers simulations of radiative-convective equilibrium over the ocean. The results were as-sorted by considering the accelerations of the particles from the time they acquired positive vertical velocity in the subcloud layer until they reached their LFC. Thus, it indicates that parcels can reach the LFC through a combination of both forces, with each dominating at different stages of the ascent. The authors concluded that mechanical forcing plays a crucial role in lifting parcels from the surface, which occurs for younger cold pools ($< 4\text{h}$), while thermodynamic forcing reduces the inhibition encountered by parcels at the convective inhibition layer, which occurs for fully developed cold pools.

Besides triggering convection, CPs are responsible for organizing it. Simulations conducted by [Feng et al. \(2015\)](#) in warm tropical oceans demonstrated that when colliding CPs, the convection formed exhibited longer lifetimes, larger spatial extents, and increased intensity compared to that generated by isolated CPs. According to the authors, the organization induced by intersecting cold pools relies on two factors: the enhanced secondary updraft velocities resulting from stronger outflows (stronger gust fronts) and due to the moister local environment being created by the CP complex.

Secondary updraft velocities are defined in this study as the upward movement on the CP edge that lifts the air, forming new convective clouds. On the other hand, the proximity of large clouds along the boundaries of intersecting cold pools, due to collision, creates a moister local environment, reducing dry entrainment and increasing the likelihood of cloud development into deep convection. This process, referred to as Moist Air Entrainment, is currently a topic of research. Figure 2.4 illustrates the authors' perspective.

The authors also highlight the role of isolated cold pools in organizing convection (Figure 2.4 (a)). In brief, as precipitation-driven downdrafts occur, moist environmental air is injected into the boundary layer outward upon contacting the surface, creating what are known as gust fronts. Subsequently, cold pools elevate this relatively moist environmental air from the surface, leading to the formation of new convective clouds along the boundaries of the cold pool. As these convective clouds develop, their growth is constrained by the entrainment of dry air from the upper boundary layer, particularly when they are small in size and the spacing between

Figure 2.4 - Cold pools organizing convection.



Source: Feng et al. (2015).

them is large.

Up to this point, we have discussed the composition of a cold pool, particularly in low wind shear environments (oceans). We have explored the mechanisms associated with cold pools that are responsible for initiating new convection and their role in organizing it. In the next section, we will begin the discussion regarding the cold pool parameterization based on previous ideas for this kind of parameterization, followed by the parameterization by Freitas et al. (2024), the subject matter of this work.

2.3 The cold pool parameterization

Over the past few years, several schemes have been developed to represent the effect of cold pools within global circulation models and convection-permitting models.

Grandpeix and Lafore (2010) introduce the concepts of Available Lifting Energy (ALE) and Available Lifting Power (ALP) to address the influence of wakes (addressed by the authors as the lifting effects created by circular cold pools) in triggering and sustaining new convective cells. The parameterization was coupled into a GCM. The continuation of this research (GRANDPEIX et al., 2010) examines two case studies within a single-column framework: one over the Sahel in West Africa and the other over the West Pacific Ocean during the TOGA COARE period. The study also investigates the sensitivity of basic parameters within this parameteriza-

tion. Additionally, a review of the earlier literature on cold pool parameterization can be provided by checking the references within this paper.

[Genio et al. \(2015\)](#) investigate the impacts of entrainment, rain evaporation, downdrafts, and cold pools with the Goddard Institute for Space Studies (GISS) Model E2 GCM. In this study, the authors modified the baseline parameterization configurations of the GCM to explore these various atmospheric phenomena. Notably, in the context of cold pools, they developed an experimental parameterization to represent weakly entraining convection in a manner that is more physically grounded compared to traditional stronger-entrainment models.

The approach conceptualizes cold pools from the perspective of the boundary layer: a less-entraining plume does not form until a time step after the entry of downdraft air into the boundary layer. This downdraft air must possess a virtual potential temperature that is cooler than the ambient air, along with a mass that constitutes at least 5% of the boundary layer mass. Upon these conditions being met, a cold pool is defined, with its depth evolution governed by the additional downdraft mass introduced in subsequent time steps and the horizontal spread rate of the cold pool. Once the downdrafts cease, a given plume remains viable if the environmental air that is lifted to the altitude of the cold pool's upper boundary becomes buoyant, persisting until the thermodynamic properties of the cold pool converge with those of the surrounding boundary layer. In the context of the Madden-Julian Oscillation (MJO), the authors identified limitations in the previous parameterization, which produced a less-entraining plume with insufficient frequency of occurrence. To address these deficiencies, they implemented enhancements to the specifications governing the downdraft and cold pool dynamics. These modifications aimed to improve the understanding of the influence these factors exert on MJO behavior.

The results indicate that the initial cold pool simulation closely resembles the output of the stronger-entrainment model; however, it exhibits several improvements, including more pronounced and realistic rainfall anomalies during the final week of the hindcast and a marginally slower propagation speed. On the other hand, simulations utilizing increased downdraft rates and enhanced cold pool dynamics yielded a degradation in MJO performance, characterized by a tendency towards faster propagation, diminished rainfall anomalies in the concluding week of the hindcast, and less suppressed conditions in the initial week centered over the central to eastern Indian Ocean.

More recently, [Hirt and Craig \(2021\)](#) designed a cold pool parameterization scheme,

named the Cold Pool Perturbation (CPP), in a convection-permitting model, that focuses on increasing the vertical velocity at the simulated gust fronts driven by cold pools. The idea behind CPP is briefly described as follows. A target vertical velocity scale for fully resolved gust fronts was derived using dimensional analysis and similarity theory, based on buoyancy gradients at cold pool boundaries. This scale provides a physically based estimate of the vertical velocity that should be present at gust fronts. The CPP scheme applies perturbations to the vertical velocity tendency in the gust front region, relaxing the model vertical velocity towards the target on a specified time scale. To confine perturbations to appropriate locations, the scheme uses local criteria based on strong virtual potential temperature gradients, vertical velocity thresholds, orographic filtering, and horizontal smoothing to identify cold pool gust fronts where the perturbations should be applied. Sensitivity tests were also performed to calibrate the parameters within the parameterization.

The results demonstrate that the CPP scheme modestly strengthens cold pool gust fronts by increasing vertical velocities at about 1 km height compared to the reference simulation. This enhancement leads to more frequent, larger, and more intense cold pools. The CPP scheme notably increases precipitation amounts, especially during the afternoon and evening when cold pools are more prevalent. Precipitation cells become stronger, larger, and more organized, often forming line structures or clusters, with these effects amplified at higher perturbation amplitudes. An additional effect of the CPP scheme is the strengthening of sea breeze fronts, which are similar to density currents and can also trigger convection. This leads to earlier and more intense convection near coastal areas, better matching radar observations. Although the CPP scheme has improved most measures for the quality of the forecast convection, the improvements are modest, and the differences to the observed properties of convection remain large according to the authors.

This parameterization is constrained to models that explicitly address cold pools and deep convection (with $dx < 10$ km). In conclusion, the authors propose alternative sources of convective initiation that, like cold pools, are not fully captured in models. Addressing these sources could potentially yield improved results, particularly when the parameterizations are coupled together.

Rooney et al. (2022) created a scheme that represents the lifecycle and effects of convective cold pools (named C-POOL). This approach is grounded in similarity modelling of gravity currents, which provides approximations for the propagation and evolution of cold pools.

Firstly, there is an identification of cold pools, characterized by primary properties such as depth scale and reduced gravity scale, which relate to the temperature deficit of the cold pool relative to the environment. The scheme is forced by the downdraught (downdraft) mass flux per unit area at cloud base, obtained from the convection scheme in the Met Office Unified Model (UM). This downdraught forcing provides the initial input to generate cold pools at the grid scale. The scheme prognoses cold pool properties such as buoyancy and depth, and models their interaction and decay over time. An ensemble average radius is used to approximate the front position of cold pools, enabling the model to function without tracking individual cold pool fronts explicitly.

The scheme allows for cold pools to propagate, interact, and merge within the model domain, which is demonstrated through idealized tests showing the evolution of cold pool properties and their collisions. The prognostic variables from the cold pool scheme feed back into the convection parameterization in the UM. These feedbacks affect convective diagnosis, initiation from the boundary layer, parcel ascent buoyancy, and entrainment rates, thereby influencing the convective lifecycle and improving the representation of tropical rainfall, especially the diurnal cycle. The scheme includes various parameters and coefficients, which have been tuned to physically reasonable values through model development and testing. Also, the scheme has been tested in global climate model configurations, showing improvements in the timing of the diurnal peak of tropical precipitation.

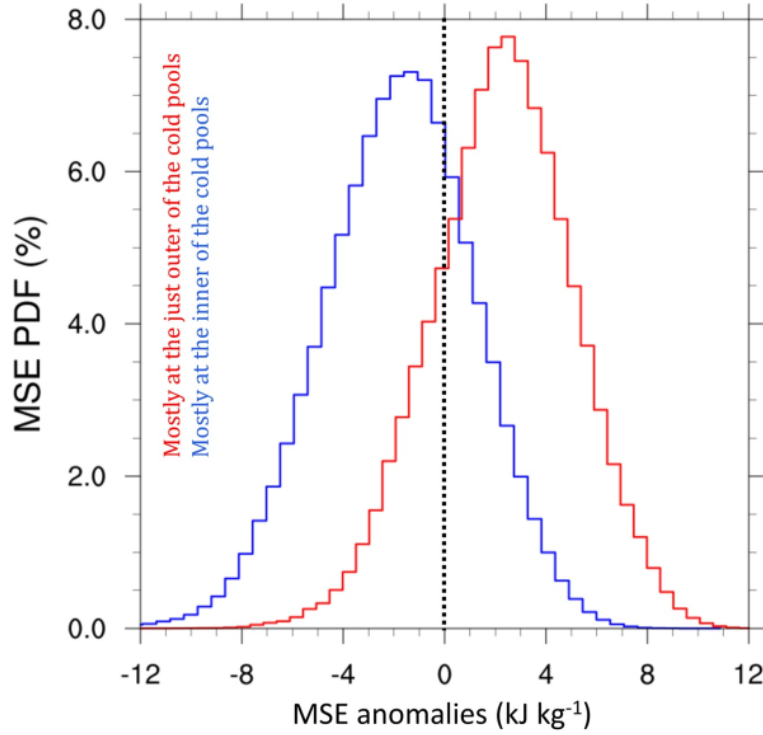
With this brief overview of cold pool parameterizations, we will now discuss the cold pool parameterization employed in this study.

[Freitas et al. \(2024\)](#), in the context of coarse-resolution atmospheric models, designed a parameterization that simplifies the cold pool phenomenon while focusing on the essential characteristics of gust front properties that trigger and organize new convection. The authors concentrated on the gust front and the associated moist rings, without explicitly considering the sub-grid regions populated by cold pools. They also noted the need for a prognostic equation for the MSE anomalies, which would help initiate stronger new convection. Additionally, they suggested that the gust front propagation should be treated diagnostically.

To better understand the environment surrounding cold pools, initial CRMs simulations were conducted. Specifically, these simulations utilized the Brazilian development on the Regional Atmospheric Modeling System (BRAMS) ([FREITAS et al., 2017](#)) over an extensive portion of the Amazon Basin. A definition of the cold pool

enclosure was established and divided into two thermodynamically distinct parts: an inner dry and cold region surrounded by moist rings. A filter was then created to highlight this notable feature, discriminating areas prone to forge (positive MSE anomalies) and inhibit (negative MSE anomalies) convection. The filter follows essentially two criteria: (i) if the near-surface vertical velocity is $> 0.1 \text{ ms}^{-1}$ and the subsequent 30 min accumulated precipitation is $> 4 \text{ mm} \Rightarrow$ may indicate a region of positive MSE deviation or the region of the moist rings and just the outer region of the cold pools; (ii) if the vertical velocity is $< -0.1 \text{ ms}^{-1}$ and the previous 30 min precipitation is $> 4 \text{ mm}$, then it may indicate a region of negative MSE deviation or the cold pool inner region (FREITAS et al., 2024). After the filter application in the dataset, figure 2.5 was produced within the whole domain and temporal range (48 hr).

Figure 2.5 - Probability Density Functions of the inner core and moist rings that compose the cold pool.



Source: Freitas et al. (2024).

By analyzing this result, one can deduce that there is a positive MSE anomaly linked to the outer regions of the cold pools, which drive new convection, while a negative anomaly corresponds to the inner parts of the cold pool, where it primarily

originates from the downdraft.

These observations prompted the authors to create a new parameter that encapsulates these MSE anomaly characteristics: a new three-dimensional positive-definite scalar field referred to as buoyancy-excess, defined as the difference between the MSE from the downdraft (H_d) and the environmental one $\tilde{H}$:

$$\beta_x = -(H_d - \tilde{H}) \quad (2.1)$$

It is assumed that the negative value of the MSE deficit (β_x) in the inner part of the cold pool approximates the MSE excess in the moist rings.

Following the narrative by Freitas et al. (2024), the next step is to provide formulations for the gust front velocity, including the MSE anomalies information, translated with the β_x parameter. According to the authors, the gust front horizontal speed is then given by:

$$V_{gf} = \kappa \left(\int_0^D \frac{1}{1 + \gamma} \frac{\beta_x}{c_p \tilde{T}} g dz \right)^{1/2} \quad (2.2)$$

where κ is the internal Froude number, $\tilde{T}$ is the environmental air temperature, g is the gravity acceleration, c_p the specific heat of dry air, z the height and γ is defined on Arakawa and Schubert (1974)³.

Further, the vertical velocity at the cold pool edge is defined as:

$$W_{gf} = \kappa \left(\int_0^D \frac{1}{1 + \gamma} \frac{\beta_x}{c_p \tilde{T}} \sin^2 \alpha g dz \right)^{1/2} \quad (2.3)$$

Here, the α represents the slope angle of the cold pool head, and the D parameter is the depth of the cold pool.

In the Freitas et al. (2024) formulation, the CP propagates only in the horizontal direction, following the terrain. To do this is defined first the mean cloud layer horizontal velocity, considering the atmospheric pressure p and the $(u, v)_{env}$ the

³For this dissertation, the equation available in Arakawa and Schubert (1974) is given by:
 $\gamma \equiv \frac{L}{c_p} \left(\frac{\partial \bar{q}^*}{\partial T} \right)_{\bar{p}}$.

horizontal environment wind:

$$(u, v)_{mcl} = \frac{1}{p_2 - p_1} \int_{p_1}^{p_2} (u, v)_{env} dp \quad (2.4)$$

Finally, the two-dimensional propagation velocity of the gust front is given by:

$$(u, v)_{prop} = (u, v)_{mcl} + \frac{V_{gf}}{|(u, v)_{mcl}|} (u, v)_{mcl} - 0.6(u, v)_{env} \quad (2.5)$$

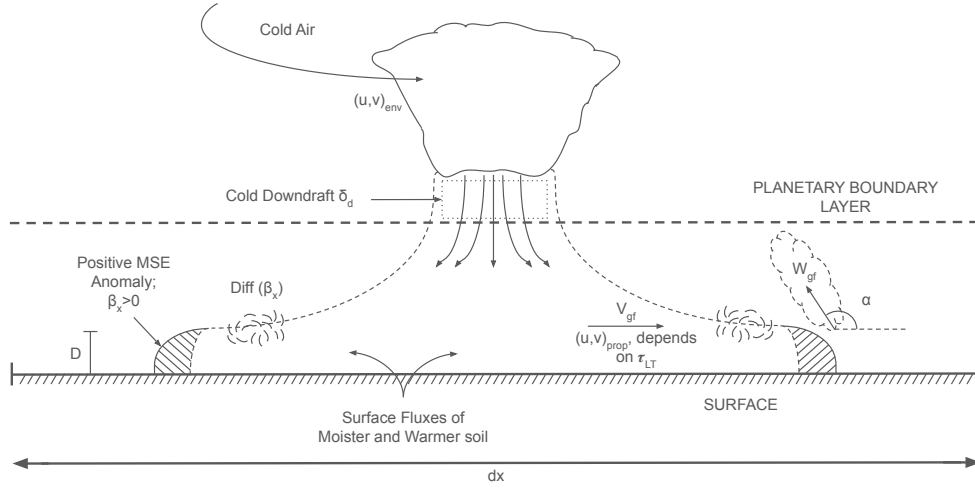
Hence, it is possible to create a prognostic equation that governs the time evolution of the MSE anomalies on the CP edges, in other words, an equation for β_x :

$$\frac{\partial \beta_x}{\partial t} = -(u, v)_{prop} \cdot \nabla \beta_x + \text{dif}(\beta_x) + \delta_d \beta_x - \frac{\beta_x}{\tau_{cp}} \quad (2.6)$$

On the right side of the equation above, the first term represents two-dimensional advection, while the second term denotes diffusion. The third term corresponds to a source primarily resulting from the downdraft detrainment mass flux (δ_d), and the final term represents a sink associated with processes other than mixing with the environment, which is calculated based on the lifetime of the cold pool. Figure 2.6 illustrates the quantities discussed above.

Note that the entire process takes place within a grid cell (dx).

Figure 2.6 - A cold pool gust front and its associated parameters.



Source: Conceptualized by the author based on Droegemeier and Wilhelmson (1987); illustrated by Arthur Schelbauer Neto (2025).

2.4 Coupling with Grell-Freitas convection parameterization

The previous parameterization was designed to be coupled within a model, not necessarily with a convection parameterization. However, for this study, we again followed the approach made by Freitas et al. (2024) and coupled it with the Grell-Freitas convection parameterization (GRELL; FREITAS, 2014; FREITAS et al., 2018).

A parameterization for convection essentially has a closure and a trigger function (EMANUEL, 1994). Therefore, the authors opted for a single closure for convection, originally proposed by Bechtold et al. (2014), which combines the stability removal formulation with the non-equilibrium diurnal cycle. The closure, which determines the updraft mass flux at the cloud base, can be stated as:

$$\text{mass flux} \propto \frac{A}{\Gamma} \quad (2.7)$$

In this context, let A represent the cloud work function and Γ denote the convective adjustment timescale. A comprehensive examination of this methodology, in

conjunction with the Grell-Freitas scheme, is presented in the work of Freitas et al. (2020). The formulation of the cloud work function, incorporating the effects of the cold pool, is now articulated as follows:

$$A = \int_{z_b}^{z_t} \frac{1}{c_p \tilde{T}} \frac{Z_u}{1 + \gamma} (H_u - \tilde{H}^*) g dz \quad (2.8)$$

As first defined by Arakawa and Schubert (1974) and introduced by Grell Grell and Dévényi (2002), the cloud work function is an integral measure of the buoyancy force, as further developed by Grell and Freitas (2014). The integral is computed from the height of the cloud base (z_b) to cloud top (z_t), and the buoyancy force here is represented by the difference between updraft and environment-saturated MSE (H_u and $\tilde{H}^*$, respectively). The gravity acceleration g , c_p the specific heat of dry air, Z_u the normalized updraft mass flux, $\tilde{T}$ the environment air temperature, and γ is the same as Equation 2.2.

The presence of the cold pools, or more specifically, the presence of the moist ring at the leading edge of the cold pools, is inside the updraft MSE:

$$H_u(z_b) = \tilde{H}(z_b) + \beta_x \quad (2.9)$$

So now, this remaining MSE at the gust front serves as an additional buoyancy source, besides the unsaturated environment MSE.

In addition to the energy available to cause air parcels to rise, the moisture at the cloud base must be defined to provide the necessary conditions for the intensification of mass flux, following this approach. This is done through the following equation:

$$q_u(z_b) = \tilde{q}(z_b) + \frac{\beta_x}{L_v} \quad (2.10)$$

Putting into words, the above equation gives us the water vapor mixing ratio $q_u(z_b)$ in the updraft at the cloud base. Note that the entire MSE available at the gust front is now expressed in the form of water vapor (see last term in the equation) and then summed with the environmental water vapor.

The second effect in the parameterization, regarding this closure approach, is the modification made at the convective adjustment time scale:

$$\Gamma = \int_{z_b}^{z_t} \frac{dz}{w_u(z)} \quad (2.11)$$

Where w_u is the sub-grid scale updraft vertical velocity. Therefore, it is directly impacted by the vertical velocity at the gust front leading edge, serving also as a boundary condition for the sub-grid scale updraft velocity:

$$w_u(z_b) = W_{gf} \quad (2.12)$$

In a nutshell, in this closure, when in tandem with the moist rings generated by the cold pools (i.e., the MSE provided and the vertical velocity at the cold pool leading edge), it can provide a stronger cloud work function and a short convective adjustment time scale, producing at the end a intense updraft mass flux at the cloud base.

Finally, a new trigger function was proposed by Freitas et al. (2024) to be incorporated into the configuration of the Grell-Freitas convection parameterization. The idea is to represent the property of gust fronts to generate new convection based on the vertical velocity at the edges of these gust fronts. Thus, in energetic terms, the kinetic energy of the air parcels associated with this phenomenon is quantified as follows:

$$E_k = \frac{1}{2} W_{gf}^2 \quad (2.13)$$

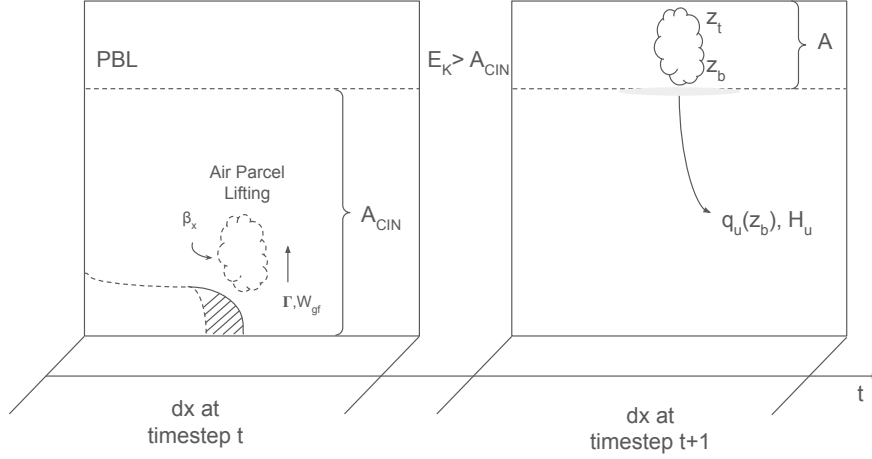
And then, the convection in a model column is triggered when:

$$E_k > |\min(A_{cin}, 0)| \quad (2.14)$$

Where A_{cin} is the cloud work function as previously defined, but now indicates the energy required for air parcels to surpass a potential convective inhibition layer that caps the Planetary Boundary Layer (PBL). This is defined as an integral from the source level of the air parcels to the cloud base.

To summarize and illustrate the aspects covered in this section, while providing a sense of continuity between the equations, Figure 2.7 was created.

Figure 2.7 - An illustration about the coupling behaviour between the cold pool parameterization and the convection parameterization described.



Source: Conceptualized by the author; illustrated by Arthur Schelbauer Neto (2025).

In brief, this parameterization conveys the energy that enables convection (both mechanically and thermodynamically) along cold pool edges. This energy is used to overcome the Convective Inhibition (CIN) barrier, allowing the formation of a new convective cell at a later time, represented in the figure as a timestep $t + 1$. The energy from the cold pools is considered necessary to trigger new convection if the condition $E_K > A_{CIN}$ is satisfied. To meet the imposed closure, the convective adjustment timescale (Γ) is defined as a function of the vertical velocity of the parcel at the gust front edge (Figure 2.7 on the left), and the cloud work function is established based on the parameters that take into account the buoyancy-excess (H_u). Additionally, moisture will be higher depending on the amount of available MSE anomaly (Figure 2.7 on the right).

2.5 Hurricanes

To comprehend why hurricanes are used as a case study for this work, it is essential to clarify key aspects of these storms. Initially (Section 2.5.1), we introduce tropical cyclones by presenting fundamental concepts, such as their classification, including the definition of the term “*hurricane*” and the processes involved in their forma-

tion. Additionally, we define key terms that are crucial for understanding the role of cold pools in these storms, particularly in terms of their structure. Finally, connecting with the broader theme of this dissertation, Section 2.5.2 reviews the existing literature that explores the relationship between tropical cyclones and cold pools.

2.5.1 Tropical cyclone basics, concepts, and structure

Tropical cyclones are intense, organized systems characterized by closed low-pressure centers that form over tropical or subtropical oceanic waters (Houze Jr, 2010; PÉREZ-ALARCÓN et al., 2021; LIN et al., 2022). These systems are referred to as *hurricanes* in the North Atlantic, *typhoons* in the North Pacific, and *cyclones* in the South Pacific and Indian Ocean basins. Despite regional naming differences, they all represent the same atmospheric phenomenon (Houze Jr, 2010; World Meteorological Organization (WMO), 2023) .

Their classification throughout the life cycle is primarily based on sustained wind intensity. A system is classified as a tropical depression when the maximum sustained wind speed⁴ is less than 63 km/h ($< 17 \text{ ms}^{-1}$). It becomes a tropical storm once this value exceeds 63 km/h ($18\text{-}32 \text{ ms}^{-1}$). When wind speeds surpass 116 km/h ($> 33 \text{ ms}^{-1}$), the system is classified as a tropical cyclone (Houze Jr, 2010; World Meteorological Organization (WMO), 2023). After reaching this threshold, the Saffir-Simpson Hurricane Wind Scale (SAFFIR, 1973; SIMPSON, 1974) is commonly used to further categorize the cyclone’s intensity. Table 2.1 presents the classification, including potential damage descriptions associated with each category. Note the addressed name “hurricanes” because it refers to a tropical cyclone happening in the North Atlantic Ocean Basin.

⁴The term maximum sustained wind speed, when applied to a particular weather system (e.g. a tropical cyclone), is the highest average wind speed measured at an unobstructed point and time at 10-m altitude during either a 1-min time span (as defined by the United States National Weather Service and the Joint Typhoon Warning Center) or a 10-min time span (as defined by the World Meteorological Organization). Source: <https://glossary.ametsoc.org/wiki/Sustained_wind_speed>

Table 2.1 - Saffir-Simpson hurricane wind scale: categories and associated damage.

Category	Sustained Winds	Types of Damage Due to Hurricane Winds
1	mph: 74–95 kt: 64–82 km/h: 119–153	Very dangerous winds will produce some damage: Well-constructed frame homes could have damage to roof, shingles, vinyl siding, and gutters. Large branches of trees will snap and shallowly rooted trees may be toppled. Extensive damage to power lines and poles likely will result in power outages that could last a few to several days.
2	mph: 96–110 kt: 83–95 km/h: 154–177	Extremely dangerous winds will cause extensive damage: Well-constructed frame homes could sustain major roof and siding damage. Many shallowly rooted trees will be snapped or uprooted and block numerous roads. Near total power loss is expected with outages that could last from several days to weeks.
3 (major)	mph: 111–129 kt: 96–112 km/h: 178–208	Devastating damage will occur: Well-built framed homes may incur major damage or removal of roof decking and gable ends. Many trees will be snapped or uprooted, blocking numerous roads. Electricity and water will be unavailable for several days to weeks after the storm passes.
4 (major)	mph: 130–156 kt: 113–136 km/h: 209–251	Catastrophic damage will occur: Well-built framed homes can sustain severe damage with loss of most of the roof structure and/or some exterior walls. Most trees will be snapped or uprooted and power poles downed. Fallen trees and power poles will isolate residential areas. Power outages will last weeks to possibly months. Most of the area will be uninhabitable for weeks or months.
5 (major)	mph: 157 or higher kt: 137 or higher km/h: 252 or higher	Catastrophic damage will occur: A high percentage of framed homes will be destroyed, with total roof failure and wall collapse. Fallen trees and power poles will isolate residential areas. Power outages will last for weeks to possibly months. Most of the area will be uninhabitable for weeks to months.

Source: Adapted from [National Hurricane Center \(NHC\) and Central Pacific Hurricane Center \(2025\)](#).

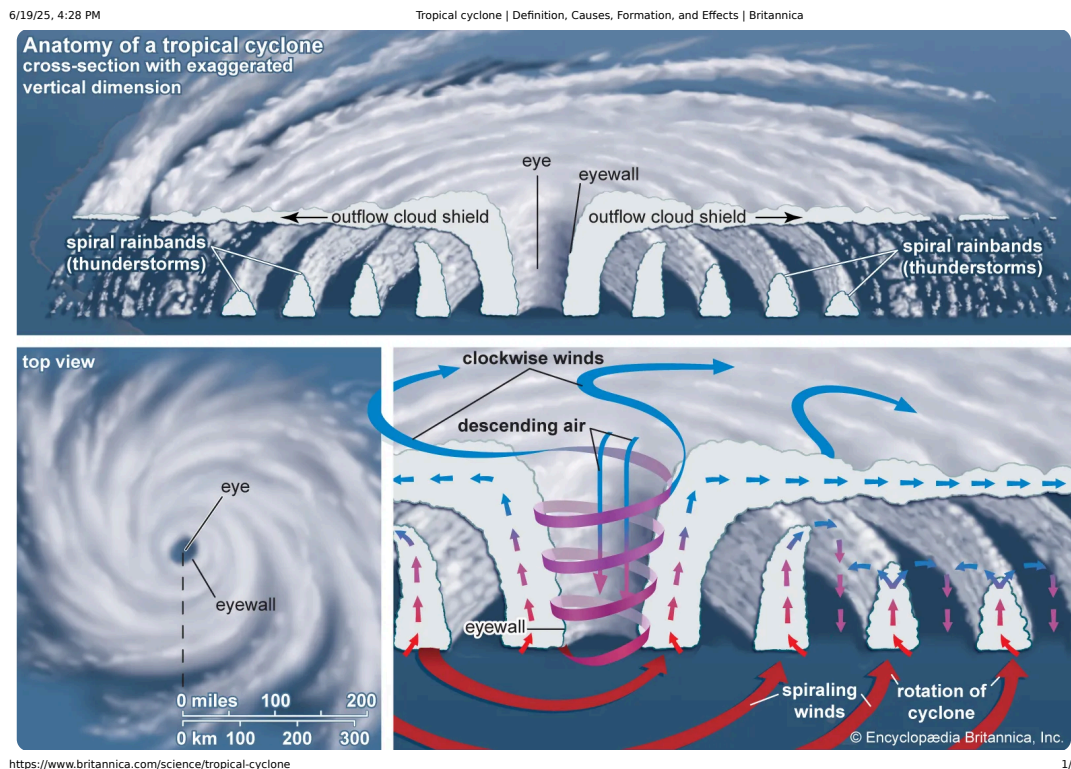
The formation of a tropical cyclone requires a delicate combination of environmental factors. According to [National Oceanic and Atmospheric Administration \(NOAA\) \(2025c\)](#), the main conditions associated with tropical cyclogenesis include sea surface temperatures of at least 27°C, extending to a depth of approximately 46 meters; an atmosphere that cools sufficiently with height, creating instability for moist convection; the presence of moist air in the mid-troposphere (around 4,900 meters); a minimum distance of about 480 km from the equator, ensuring sufficient Coriolis force; a pre-existing near-surface disturbance, such as an easterly wave; low vertical wind shear (less than 37 km/h) between the surface and upper troposphere.

These conditions collectively create an environment conducive to the organization and intensification of deep convection, necessary for a cyclone to develop. However, even when all these factors are present, not all disturbances evolve into tropical

cyclones, indicating that the genesis process still involves complex and not fully understood dynamics (PÉREZ-ALARCÓN et al., 2021).

Once developed, tropical cyclones exhibit a relatively symmetric and coherent structure when observed in mature stages. As succinctly articulated by Houze Jr (2010) and comprehensively examined by Lin et al. (2022), the system typically consists of a warm core low-pressure center at the surface; a clearly defined eye, characterized by subsidence and clear skies; an intense eyewall, where the strongest winds and deepest convection occur; spiral rainbands, extending outward from the center, containing a mix of convective and stratiform precipitation. Figure 2.8 illustrates the canonical structure of a mature tropical cyclone.

Figure 2.8 - Illustration of a hurricane structure.



Source: Zehnder (2025).

Vertically, tropical cyclones extend through most of the troposphere and exhibit strong upward motion in the eyewall, counterbalanced by subsiding air in the eye. Inflow at low levels converges toward the center and rises in the eyewall, while

outflow diverges near the tropopause (LIN et al., 2022).

Outside the core region, spiral rainbands play an essential role in transporting heat, moisture, and momentum throughout the cyclone. These bands are organized into inner and outer rainbands based on their radial distance from the storm center (BARNES et al., 1983; LIN et al., 2022).

Rainbands are composed of both convective cells and stratiform regions, often aligned in alternating patterns. Convective updrafts dominate in active cells, whereas downdrafts, especially those driven by evaporative cooling, occur within decaying convection and stratiform areas. These downdrafts can generate the cold pools. According to Sawada and Iwasaki (2010a), Sawada and Iwasaki (2010b), the cold pools can modify the local environment and influence the triggering of new convection within the rainbands. This behavior creates a dynamic feedback loop between downdraft-induced cold pools and the organization of convection, which plays a significant role in the structure and evolution of the tropical cyclone. As will be discussed in the following section, these interactions are fundamental to understanding propagation and intensification mechanisms within the cyclone and represent an active area of research.

2.5.2 The role of cold pools in tropical cyclones

Although cold pools have been widely studied in the context of both midlatitude and tropical convective systems, their influence within tropical cyclones (TCs), particularly mature hurricanes in the North Atlantic basin, remains underexplored. As highlighted in the literature review presented by Sawada and Iwasaki (2010b), relatively few observational studies have documented cold pool behavior inside TCs, due to difficulties in capturing a three-dimensional structure of cold pools. Among the observational efforts to study this phenomena, Eastin et al. (2012) offered a detailed analysis of common kinematic and thermodynamic cold pool structures in the outer rainbands of Tropical Storm Hanna (2008), suggesting “that cold pool formation, structure, and evolution depend on several multiscale processes, which can be difficult to simultaneously observe”, alongside cold pool effects within the rainband organization and longevity.

Given that background, idealized simulations are needed to investigate such a process. While these studies lack the complexity of real-case scenarios, they provide valuable insights into the possible mechanisms by which cold pools interact with tropical convection and vortex dynamics.

For instance, [Sawada and Iwasaki \(2010b\)](#) conducted idealized numerical simulations to explore the formation and evolution of rainbands in tropical cyclones, emphasizing the role of cold pool dynamics. Their results indicate that the rainband characteristics observed in their simulations are more strongly governed by cold pool processes than by mechanisms proposed in earlier studies. Specifically, the development of rainbands is initiated through upstream convective cell generation (a parcel that was ascending vertically due to vertical wind at the cold pool leading edge), triggered by interactions between cold pools and low-level inflow regions with high equivalent potential temperature. This mechanism leads to the formation of spiral-shaped rainbands that propagate radially outward, aligned with the azimuthally averaged streamlines extending from the surface up to the top of the PBL. These behaviors closely resemble observed radar echo patterns, according to the authors. Furthermore, the authors argue that cold pool dynamics may have implications not only for storm structure but also for improving forecasts of tropical cyclone tracks and intensity.

In a similar context, [Cai and Tang \(2019\)](#) investigated cold pools located just outside the eyewall of tropical cyclones and their role in the formation of inner rainbands. Using idealized simulations, the study revealed that these eyewall cold pools differ markedly from those typically observed in outer rainbands. They are generated by the evaporation of precipitation falling from the eyewall into a surrounding dry air layer, resulting in arc-shaped cold pools that are often larger and more persistent than those in the outer core. These cold pools rotate cyclonically around the storm center, with varying shapes and intensities, and propagate more slowly than the background low-level flow. Importantly, the authors propose a new thermodynamic and dynamic mechanism for inner rainband formation: as the cold pool interacts with the low-level inflow, it decelerates the flow, enhances convergence, and induces upward motion at its leading edge, ultimately triggering new convection and the development of inner rainbands. Moreover, the study identifies a relationship between the intensity of the eyewall cold pool and the overall intensity of the tropical cyclone, indicating that “when the TC has moderate strength, the eyewall cold pool is well defined and strong; it then weakens as the TC intensifies”.

Conversely, in the idealized simulations conducted by [Grant et al. \(2018\)](#), which examined cold pool dynamics within tropical oceanic convective systems, the authors found that weakening cold pools led to more intense convection. This result was explained through the use of passive tracers, which revealed that strong cold pools introduce negatively buoyant air into developing updrafts, thereby reducing

their strength through entrainment. When cold pools are weaker, this negative buoyancy effect is diminished, allowing updrafts to intensify. Interestingly, despite the increased convective strength, the simulations showed that the propagation speed and mesoscale organization of the systems were largely insensitive to the strength of the cold pools. Instead, the study identified gravity waves as the primary modulators of system propagation and mesoscale structure. According to the authors, these findings suggest that while cold pools can influence convective intensity, other processes, such as gravity wave dynamics, may play a more dominant role in shaping the organization and movement of tropical convective systems.

Moving beyond idealized experiments, [Chen et al. \(2014\)](#) used the Weather Research Forecasting Model (WRF) model in combination with radar observations to investigate the effects of cold pools under more realistic conditions during Typhoon Fengshen (2008). The study identified a strong cold pool associated with a large stratiform precipitation region, which played a crucial role in maintaining ongoing convective development. According to the authors, this process occurs when warm and moist air with high equivalent potential temperature converges with the leading edge of the cold pool. This interaction generates low-level convergence and upward motion, favoring the continuous formation of new convective cells. As a result, the presence and persistence of the cold pool contribute to prolonging the lifetime of the outer mesoscale convective system, making it longer-lasting than the typically shorter-lived distant rainbands of the tropical cyclone.

These findings collectively suggest a significant, though under-researched, role of cold pools in the evolution of tropical cyclone structure and intensity. However, most existing studies either depend on idealized environments or focus on isolated case studies, underscoring the necessity for further investigation within more generalized or climatological contexts. To fill this gap, the current study explores the impact of cold pools on two North Atlantic hurricanes using high-resolution simulations from a GCM. By examining the characteristics of cold pools and their interactions with convective structures in a controlled yet realistic framework, this research seeks to enhance our understanding of the mesoscale processes that govern tropical cyclone behavior.

2.6 Key points recall

This section summarizes the fundamental concepts and mechanisms underlying buoyancy-driven convection and the role of cold pools in convective processes, particularly within tropical systems.

We began by defining buoyancy as the driving force for convection, arising from temperature (or virtual temperature) gradients. Buoyancy is the force behind the vertical transport of heat, moisture, and momentum, making it essential for understanding cloud formation. A key metric used to quantify the convective potential of an environment is the Convective Available Potential Energy (CAPE). Deep convection typically arises when CAPE is high and Convective Inhibition (CIN) is low, or when external triggers are present to release the buoyant energy.

Cold pools form when precipitation partially evaporates, cooling the sub-cloud air and creating a negatively buoyant air mass. Initially studied as density currents, cold pools were thought to trigger new convection mechanically via gust front lifting. However, [Tompkins \(2001\)](#) challenged this view by showing that in low-shear oceanic environments, thermodynamic forcing, particularly moisture anomalies at the edges of CPs, plays a dominant role in convection initiation by reducing CIN.

[Tompkins \(2001\)](#) further described cold pool evolution in three stages: (i) formation, via evaporation-induced cooling beneath the convective towers; (ii) maturity, characterized by strong downdrafts injecting dry air into the core and moist bands along the CP edges; and (iii) decay, during which surface fluxes and entrainment begin to restore temperature and moisture, especially at the periphery, where CIN is lowered and CAPE is maximized, leading to ideal conditions for new convective cells.

This then led to the conceptual division between the core and the edge of cold pools, with distinct thermodynamic properties. While the origin of moisture anomalies in the rings remains debated, whether from evaporation, surface fluxes, or remnants of previous CPs, it is agreed that both mechanical and thermodynamic forcings contribute to convection, depending on the CP's age and structure ([TORRI et al., 2015](#); [FENG et al., 2015](#)).

Several parameterization efforts have attempted to represent CP dynamics in atmospheric models. Studies by [Grandpeix and Lafore \(2010\)](#), [Genio et al. \(2015\)](#), and others incorporated both mechanical and thermodynamic effects of cold pools and the gust front propagation. These schemes require careful calibration and have improved the representation of key phenomena like the diurnal cycle of precipitation.

In particular, [Freitas et al. \(2024\)](#) developed a novel parameterization aimed at capturing essential cold pool characteristics, notably the gust front and its associated moist rings. This scheme is the subject matter of this work. The scheme diagnoses

cold pools using a new quantity called buoyancy-excess, computes horizontal and vertical winds at the cold pool edge, and models the gust front propagation using a prognostic equation that incorporates mean cloud layer winds and environmental damping. Buoyancy-excess itself evolves through advection, diffusion, downdraft forcing, and a sink term defined by a cold pool lifetime parameter. The last parameter will be tuned within our experiments. This scheme was coupled with the Grell-Freitas convection scheme, modifying the cloud work function, cloud base moisture mixing ratio, convective adjustment time scale, and adding a new trigger based on kinetic energy commanded by cold pools.

Finally, section 2.5 examined the role of cold pools within tropical cyclones (TCs). While cold pools have been extensively studied in midlatitude and tropical convection, their behavior within TCs remains relatively unexplored due to observational challenges. However, several idealized modeling studies (e.g [Sawada and Iwasaki \(2010a\)](#), [Cai and Tang \(2019\)](#)) and some real-case analyses [Chen et al. \(2014\)](#) have revealed that cold pools influence both outer and inner rainband dynamics.

Key findings include (i) cold pools can trigger new convection in rainbands via convergence and uplift at the gust front; (ii) eyewall cold pools, formed by rain evaporation into dry surrounding air, may rotate cyclonically and influence the structure and intensity of the TC; (iii) the strength and lifetime of cold pools appear linked to TC intensity and rainband organization; and (iv) in some cases, weaker cold pools were associated with stronger convection due to reduced negative buoyancy in developing updrafts ([GRANT et al., 2018](#)).

However, much of the existing research is still confined to idealized frameworks or isolated instances. To address this limitation, the current study explores cold pool behavior in two North Atlantic hurricanes, utilizing GCM simulations to enhance our understanding of how cold pools influence convective processes in actual tropical cyclone environments.

3 DATA AND MODEL

3.1 Data

This section presents the observational datasets used as a reference for evaluating the model forecasts. The selected datasets, described in detail in Sections 3.1.1 to 3.1.6, are commonly employed in tropical cyclone studies and provide reliable information on storm trajectory, intensity, and rainfall. They have been used in several studies (ZHOU; MATYAS, 2021; BOPAPE et al., 2021; MAZZA; CHEN, 2023; DULAC et al., 2024; YANG et al., 2024; MAY et al., 2024) and support a robust assessment of the model’s performance.

3.1.1 IBTrACS

The International Best Track Archive for Climate Stewardship (IBTrACS) (KNAPP et al., 2010) offers comprehensive data on the location and intensity of global tropical cyclones. It emphasizes parameters such as geographic position (latitude and longitude), maximum sustained wind speed (measured in knots), and minimum central pressure (in millibars). A complete list of variables and their definitions can be found at the provided link¹ This dataset features a spatial resolution of 0.1° (approximately 10 km) and primarily reports data at a temporal resolution of 6 hours, although it can be interpolated to intervals of 3 hours; for our purposes, we will be utilizing the 6-hour data. The dataset covers the period from 1841 to the present, allowing users to explore it by searching for specific oceanic basins, including the North Atlantic, South Atlantic, Eastern North Pacific, Western North Pacific, South Pacific, South Indian, and North Indian Oceans.

The dataset is built upon information from various agencies. According to the documentation, the sources include all available resources utilized by forecasters, such as surface observations, aircraft reconnaissance flights, and satellite observations.

Hereafter, we will refer to this dataset as simply the best track dataset.

3.1.2 ERA5

ERA5 represents the fifth generation of atmospheric reanalysis conducted by the European Centre for Medium-Range Weather Forecasts (ECMWF), serving as a vi-

¹NOAA - National Centers for Environmental Information. **IBTrACS version 4r01: technical details**. Asheville, NC: NOAA NCEI, abr. 2025. 31p. (PDF File). Available at: <https://www.ncei.noaa.gov/sites/g/files/anmtlf171/files/2025-04/IBTrACS_version4r01_Technical_Details.pdf> Accessed on: June 27, 2025.

tal resource for both global climate and weather studies. This reanalysis integrates a forecast model with an advanced data assimilation scheme. Specifically, ERA5 employs the ECMWF Integrated Forecast System (IFS) CY41R2 forecast model, executing twice-daily short-term forecasts (18 hours) derived from analyses conducted at 06:00 and 18 UTC. Check the link ² for a comprehensive understanding of the physical parameters encompassed within the model. The 4D-Var data assimilation method assimilates a diverse array of observations, including satellite data, ground station measurements, and reconnaissance aircraft information. This process utilizes 12-hour time windows from 09 UTC to 21 UTC and from 21 UTC to 09 UTC (the subsequent day).

The initial configuration of the product involves 137 hybrid sigma/pressure levels in the vertical, with the uppermost level being at 0.01 hPa, and it maintains a horizontal resolution of 0.28125° (approximately 30 km). Atmospheric data are accessible across interpolated 37 pressure levels and at a horizontal resolution of 0.25° . Consequently, the dataset comprises four primary subsets for download: hourly and monthly products, available on pressure levels (restricted to 37 levels) and single levels that encompass atmospheric, ocean-wave, and land surface quantities.

Reanalyses are not constrained by the necessity for timely forecasts, which allows for an extended period to collect observations. Furthermore, when assessing historical data, enhanced versions of the original observations can be integrated, thereby enhancing the quality of the reanalysis product. As a result, reanalyses provide a physically and dynamically consistent global representation of the atmospheric state at each time step. The major advantage of atmospheric reanalysis, particularly in the context of studying tropical cyclones, lies in its capacity to facilitate the analysis of the internal three-dimensional structure of contemporary TCs, alongside the large-scale environmental conditions surrounding them (DULAC et al., 2024).

The ERA5 data is accessible via the Climate Data Store and downloadable through the Climate Data Store (CDS) Application Program Interface (API). The variables under investigation are highlighted in Table 3.1.

²European Centre for Medium-Range Weather Forecasts. **IFS documentation CY41R2 – Part IV: Physical processes**. Reading, UK: ECMWF, 2016. 213 p. [PDF file]. Available at: <https://www.ecmwf.int/en/elibrary/79697-ifs-documentation-cy41r2-part-iv-physical-processes>.> Accessed on: June 27, 2025.

Table 3.1 - ERA5 atmospheric variables selected for model evaluation.

Variable name	Long name	Unit
Msl*	Mean sea level pressure	Pa
I10fg*	Instantaneous 10 metre wind gust	ms ⁻¹
tp*	Total precipitation	m
sst*	Sea Surface Temperature	K
U**	U component of wind (Zonal Wind)	ms ⁻¹
V**	V component of wind (Meridional Wind)	ms ⁻¹

Legend: * single levels; ** pressure levels.

Source: Made by the author (2025).

3.1.3 GPM-IMERG

NASA’s Integrated Multi-satellite Retrievals for GPM (Global Precipitation Measurement) (IMERG) (HUFFMAN, 2019) algorithm synthesizes information from the GPM satellite constellation. Satellite data are particularly valuable to get information in areas of the Earth where ground-based precipitation-measuring instruments are limited, such as the Atlantic’s Main Development Region.

According to the Technical Documentation³, estimates from various precipitation-relevant passive microwave (PMW) sensors within the GPM constellation are processed using an algorithm. These estimates are then gridded, intercalibrated with the GPM Combined Radar Radiometer Analysis product, and integrated into half-hourly fields with a horizontal resolution of $0.1^\circ \times 0.1^\circ$, covering latitudes from 60°S to 60°N . This data is provided to the Climate Prediction Center (CPC) Morphing-Kalman Filter (CMORPH-KF) quasi-Lagrangian time interpolation procedure, as well as undergoing a re-calibration that applies to the Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks (PERSIANN) Dynamic Infrared–Rain Rate (PDIR) infrared precipitation retrievals.

The product is offered in three stages: Early Run, Late Run, and Final Run. Researchers are encouraged to utilize the Final Run data for comprehensive analyses, as this stage incorporates monthly gauge data to create research-level products. However, it is important to note that the Final Run data has a latency period of approximately 3.5 months from the time of observation. Accordingly, we will employ

³NASA – National Aeronautics and Space Administration. **IMERG Technical Documentation**. Greenbelt, MD: NASA GES DISC, 2020. 97 p. [PDF file]. Available at: https://arthurhou.pps.eosdis.nasa.gov/Documents/IMERG_TechnicalDocumentation_final.pdf. Accessed on: June 27, 2025.

the Final Run product, specifically using the NASA Goddard Earth Sciences (GES) Data and Information Services Center (DISC) V07 data, accessible through the link ⁴, which offers precipitation data in netCDF format.

Lastly, concerning temporal distribution, this dataset encompasses the interval from June 2000 to September 2024, as of the time of this analysis, being available for download every half-hour, daily, or monthly. For our analysis, we intend to concatenate the data to create hourly aggregates from the half-hourly product, which will subsequently be extrapolated to a resolution of approximately 30 km (our forecast resolution).

3.1.4 GPM-MERGIR

In the context of convective cloud modeling, it is essential to evaluate the model's capability to accurately represent cloud morphology. This assessment can be achieved through the analysis of Infrared (IR) Brightness Temperature (Tb), also known as equivalent blackbody temperature, using satellite data. The observational dataset we are currently employing comes from the NASA GES DISC GPM Merged 4-Km IR Tb dataset (GPM-MERGIR) (JANOWIAK et al., 2017), which is sourced from the NOAA Climate Prediction Center/National Centers for Environmental Prediction/National Weather Service (CPC/NCEP/NWS). The satellites contributing to this dataset include the Geosynchronous Operational Environmental Satellites (GOES) from the United States, the Geosynchronous Meteorological Satellite (GMS), followed by the Multi-functional Transport Satellite (MTSat) and Himawari from Japan, as well as the Meteorological Satellite (Meteosat) from the European Community, which forwards IR imagery to the CPC. A complete list of these satellites can be found at this link⁵.

The data is made available periodically in half-hour increments, covering latitudes from 60°S to 60°N with a pixel resolution of 4 km, dating back to 1 January 1998. In addition to direct downloads of netCDF-4 format data, GES DISC also provides data in binary, ASCII, and netCDF-3 formats via the OPeNDAP interface.

⁴NASA – National Aeronautics and Space Administration. **Precipitation Data Directory**. Greenbelt, MD: NASA GES DISC, 2025. Available at: <<https://gpm.nasa.gov/data/directory>> Accessed on : June 16, 2025.

⁵NOAA/NWS/CPC. **Merged 4 km IR Tb Dataset Documentation**. Washington, DC: NOAA/NCEP, Apr. 17, 2025. 1 p. [PDF file]. Available at: <<https://docserver.gesdisc.eosdis.nasa.gov/public/project/GPM/CPC-4kmIR-Sats.pdf>> Accessed on: June 27, 2025.

3.1.5 GSMaP

Another satellite-based combined microwave-IR precipitation dataset will be utilized to evaluate precipitation generated by tropical cyclones. The Global Satellite Mapping of Precipitation (GSMaP)⁶, supported by the JAXA Precipitation Measuring Mission (PMM) Science Team, offers a multi-satellite global precipitation map as part of the Global Precipitation Measurement (GPM) Mission. It employs the Dual-frequency Precipitation Radar (DPR) onboard GPM core satellites, along with other GPM constellation satellites and geostationary satellites (website). Additionally, in the GSMaP products, apart from GSMaP_NOW, the globally merged, full-resolution ($\sim 4\text{km}$) infrared data produced by NOAA/CPC has also been utilized⁷.

The key feature of the GSMaP algorithm is its use of various attributes derived from the spaceborne precipitation radar, including TRMM/PR and GPM/DPR. This algorithm generates a rainfall rate product (in mm/hr) that covers a global extent (60°N to 60°S) with a horizontal resolution of 0.1° (latitude/longitude) and is available on an hourly basis.

The differences, features, and performance of this dataset compared to GPM-IMERG in tropical cyclone cases are widely discussed in the literature (REDDY et al., 2022; BAGTASA, 2022; YANG et al., 2024).

3.1.6 NCEP/EMC Stage IV

Hurricane Helene (one of the case studies that will be explored) resulted in significant rainfall across the continental United States, particularly in North Carolina, as we will detail later. Our observations (available for review in the Appendix) indicate that GPM-IMERG did not accurately capture these extreme conditions. Consequently, we opted to use this dataset to evaluate model-simulated rainfall, specifically in the United States continental region. For this purpose, we utilized the Stage IV hourly precipitation dataset produced by the Environmental Modeling Center (EMC) of the National Centers for Environmental Prediction (NCEP), United

⁶Westen, D. Asymmetry of AMOC Hysteresis in a State-Of-The-Art Global Climate Model. **Geophysical Research Letters**, v51, n4, p 1-9, 2023. Available at: <<https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2023GL106414>> Accessed on: June 27, 2025.

⁷Kubota, T., K. Aonashi, T. Ushio, S. Shige, Y. N. Takayabu, M. Kachi, Y. Arai, T. Tashima, T. Masaki, N. Kawamoto, T. Mega, M. K. Yamamoto, A. Hamada, M. Yamaji, G. Liu and R. Oki. Global Satellite Mapping of Precipitation (GSMaP) products in the GPM era, **Satellite precipitation measurement**, v1, n67, pp 355-373, 2020, Available at: <https://doi.org/10.1007/978-3-030-24568-9_20> Accessed on: June 27, 2025.

States. The Stage IV product is a multi-sensor, gridded analysis derived from the regional “Stage III” precipitation estimates produced by the 12 River Forecast Centers (RFCs) that cover the Continental United States (CONUS). These regional mosaics are manually quality-controlled by each RFC and then merged into a national-scale product by NCEP.

Stage IV differs from its predecessor, Stage II, primarily due to the inclusion of manual quality control procedures at the RFC level. While radar reflectivity provides spatial coverage and high resolution, the integration of gauge observations ensures ground-truth reliability. However, certain limitations remain. For example, during winter storms, many stations lack the equipment to melt frozen precipitation, resulting in underestimated liquid equivalents and potentially erroneous low values in the dataset.

The Stage IV data are available at hourly, 6-hourly, and 24-hourly intervals (the latter accumulated from the 6-hourly data) and are distributed on a 4-km polar-stereographic HRAP grid (1121 x 881 grid points). Since July 20, 2020, the data format has transitioned from GRIB1 to GRIB2 (*.grb2)⁸, which should be prioritized as the GRIB1 files may be incomplete. We will use the hourly product in this work.

Temporal coverage begins on 31 December 2001 and continues through the present, with the inclusion of Alaska and Puerto Rico data starting in April 2017. It is important to note that there are documented instances of missing or incomplete data for certain dates or regions, especially during heavy precipitation events or in complex terrain.

Due to its incorporation of radar-based spatial continuity and gauge-based validation, Stage IV is recognized as a key observational benchmark for model verification and hydrometeorological studies. The dataset is publicly available through the National Center for Atmospheric Research (NCAR) Research Data Archive (ds507.5) and the National Water Prediction Service. It was provided for this work at the Helene period as a courtesy by Haiqin Lin.

In the literature review, comparisons can be found between this type of dataset and others, such as satellite data and ERA5 products, particularly concerning landfalling tropical cyclones in the United States. For example, [Mazza and Chen \(2023\)](#) offer an overview of these datasets within the context of tropical cyclone climatology

⁸The GRIB2 dataset can be converted into a NETCDF4 file by using the `sdfwrite` function inside the Grid Analysis and Display System (GrADS)

and conclude that NCEP/EMC Stage IV (hereafter referred to simply as NCEP) identifies significantly higher precipitation rates in landfalling tropical cyclones than the other datasets.

3.2 Numerical model: MONAN

The model utilized for this study is part of the next generation of numerical models currently under development. The Model for Ocean-LaNd-Atmosphere Prediction (MONAN) is an advanced project by the Brazilian Centre of Weather and Climate Prediction (CPTEC) to become the new standard for weather and climate forecasting, nowcasting, and hindcasting. MONAN is an adaptation of the Model for Prediction Across Scales (MPAS), and further details will be provided in the subsequent paragraphs.

The MONAN is a collaborative initiative led by the National Institute for Space Research (INPE) and the Ministry of Science, Technology, and Innovation (MCTI). Its primary objective is to improve weather and climate forecasting in Brazil, South America, and the Caribbean at all spatial and temporal scales, particularly in climate change scenarios.

MONAN employs an MPAS dynamical core, supplemented by various physics innovations developed by the scientific community. The most recent version of MONAN is 1.0.0 (last seen on July 2, 2024) with ongoing enhancements available for review in the link⁹. The present version exhibits differences from the MPAS framework, particularly in its physical bulk. Specifically, MONAN has integrated the Grell-Freitas Cold Pool parameterization Freitas et al. (2024) as well as the Simplified Higher Order Closure - Mass Flux (SHOC-MF) planetary boundary layer (PBL) scheme, which is currently under implementation by Guilherme Farache.

Regarding the physics suite, MPAS/MONAN offers two configurations: “Mesoscale Reference” and “Convection-permitting”, which vary based on resolution. The Mesoscale Reference is suitable for mesoscale resolutions (greater than 10 km cell spacing; i.e., $dx > 10$ km) but, as noted in the MPAS manual¹⁰, is not ideal for convective-scale simulations since the Tiedtke scheme can eliminate convective in-

⁹INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS (INPE). **Model for Ocean-laNd-Atmosphere predictioN (MONAN)**. 2025. Available at: <<https://monanadmin.github.io/>> Accessed on: June 27, 2025.

¹⁰NCAR - National Center for Atmospheric Research, Mesoscale & Microscale Meteorology Laboratory. **MPAS-Atmosphere Users’ Guide 8.3.0**. Boulder, CO: [s.n.], 2025. 233p [PDF file]. Available at: <https://www2.mmm.ucar.edu/projects/mpas/mpas_atmosphere_users_guide_8.3.0.pdf> Accessed on: June 27, 2025.

stability before the resolved-scale motions (convective cells) can effectively respond. On the other hand, the Convection-permitting option is designed for spatial resolutions that accommodate both explicitly resolved hydrostatic and nonhydrostatic motions. This suite is recommended for any MPAS applications employing convection-permitting meshes ($dx < 10$ km), including variable-resolution meshes that span hydrostatic to nonhydrostatic resolutions.

For our simulations ($15 \text{ km} < dx < 60 \text{ km}$), the current physics parameters are summarized in Table 3.2:

Table 3.2 - Physics bulk configuration

Parameterization	Mesoscale reference (30 km)
Microphysics	WSM-6
Convection	Grell-Freitas MONAN
Boundary layer (BL)	MYNN
Gravity wave drag by orography	YSU
Longwave radiation (LW)	RRTMG
Shortwave radiation (SW)	RRTMG
Cloud fraction for radiation	Cloud Fraction Monan
Surface layer	MYNN
Land surface	Noah

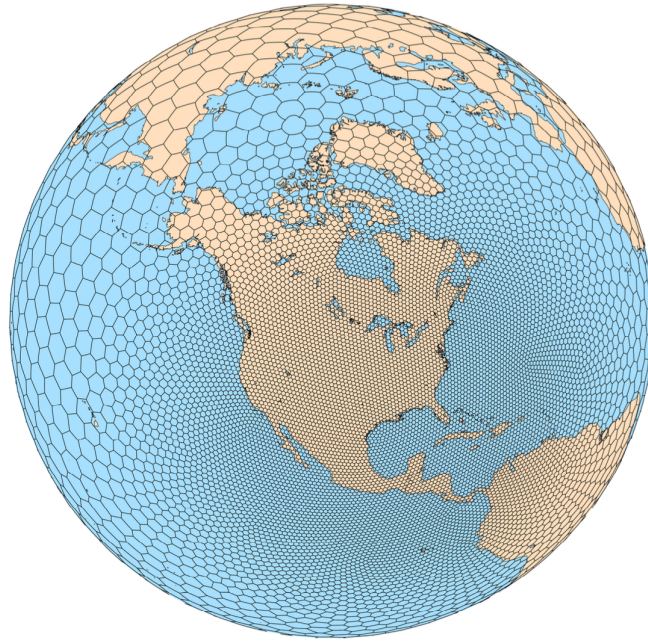
Source: Made by the author (2025).

The cloud microphysics utilizes the single-moment 6-class scheme as described by [Hong and Lim \(2006\)](#). The convection parameterization, based on the mass flux approach developed by [Grell and Freitas \(2014\)](#) and [Freitas et al. \(2018\)](#), is implemented here along with the cold pool scheme proposed by [Freitas et al. \(2024\)](#). Both the Planetary Boundary Layer (PBL) and Surface Layer parameterizations employ the Mellor-Yamada-Nakanishi-Niino level 2.5 closure Turbulent Kinetic Energy (TKE) based scheme ([NAKANISHI; NIINO, 2009](#)). Orographic gravity-wave drag is represented using the Yonsei University (YSU) PBL scheme (YSU - ([HONG, 2010](#))). For solar (shortwave) and terrestrial (longwave) radiative transfers, the Rapid Radiative Transfer Model for GCMs ([IACONO et al., 2008](#)) radiation scheme is utilized. The Cloud fraction for radiation used is entitled Cloud Fraction MONAN and can be checked in [Chaboureau and Bechtold \(2005\)](#) studies. Finally, the Land Surface parameterization is based on the Noah community model ([MITCHELL, 2005](#)).

All simulations were conducted globally, utilizing a uniform horizontal grid spacing

dependent on the specific experiment, which was set at 15 km, 30 km, and 60 km. The model has 55 vertical levels, set at depths of 0 m and 30000 m (30 km), with the ocean as the reference. Due to the global scope, only a single initial condition was required, sourced from the ERA5 and Global Forecast System (GFS) models. The duration of the model time integration varies according to the experiment and will be detailed in the results section, along with the initial time of integration. Figure 3.1 displays a map generated with MPAS-A and Figure 3.2 a map of convective rain accumulated over 2 days forecasted by MONAN.

Figure 3.1 - MPAS-A meshes and the ability to configure the meshes at different resolutions.



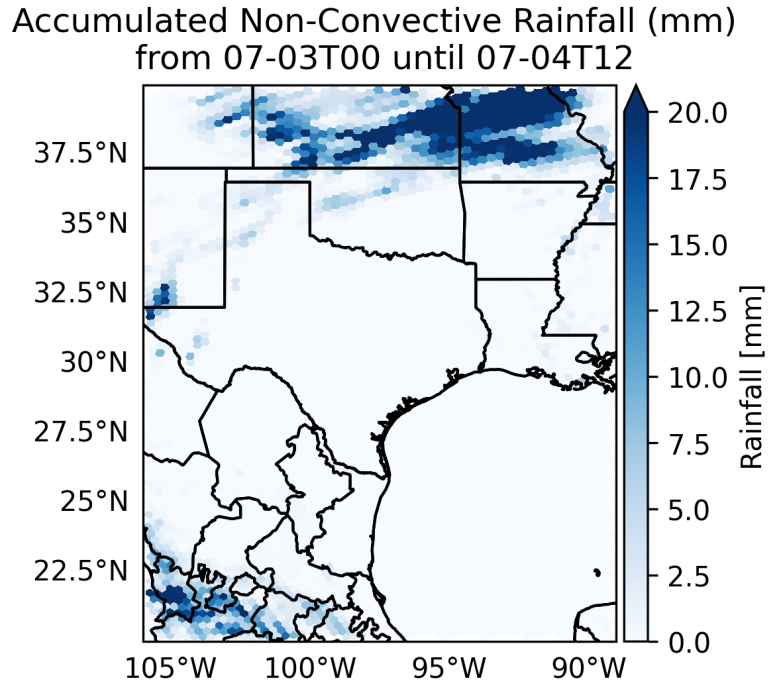
Source: NSF NCAR/MMM (2024).

As one can see, one of the big differences between MPAS and its predecessor, the WRF model, is that the model is discretized on centroidal Voronoi meshes using a C-grid staggering of the prognostic variables, allowing variable horizontal resolution, and also solving the equations of motion directly on these unstructured meshes (SKAMAROCK et al., 2012).

For our study, a set of variables was chosen to perform it. Table 3.3 indicates the variable name inside MONAN, alongside with the long name and the unit.

Total rainfall was computed by simply summing the variables *rainnc* and *rainc* for

Figure 3.2 - 30km horizontal resolution at native mesh in MONAN.



Source: Made by the author (2025).

Table 3.3 - MONAN Atmospheric Variables Selected for Model Evaluation

Variable name	Long name	Unit
rainc	Convective rainfall	mm/h
rainnc	Not Convective rainfall	mm/h
mslp	Mean sea level pressure	hPa
wspd	Wind speed	m/s
ctt	Cloud top temperature	°C

Source: Made by the author (2025).

each latitude/longitude point. The *wspd* is the module of the squared sum of zonal and meridional winds at the first model level.

3.2.1 Initial condition generation

To begin the integration process, Initial Conditions (IC) must be prepared by MPAS requirements, which can be found at the footnote of this page¹¹. We generated the initial conditions using the Weather Research & Forecasting Model (WRF) Pre-Processing System (WPS), utilizing data obtained from ERA5, which we pre-processed with the WPS. These initial conditions were then uploaded into the MONAN IC folder and executed accordingly. For the IC sensitivity test, we also acquired a GRIB file from the Global Forecast System (GFS) model, a courtesy provided by Saulo R. Freitas, who already had the data available. The model will initially be set up with a global 30 km MPAS grid, with plans to modify this in future sections. It is important to note that a new IC will need to be created for each resolution.

3.2.2 Post-processing

Since the model output is generated on a non-structured grid, a post-processing step is necessary to map the native MPAS output to other meshes, enabling visualization on a standard latitude/longitude map. To achieve this, MONAN employs the convert-MPAS project. This approach utilizes a nearest-neighbor scheme to remap integer fields to the target grid, with additional information available at this link¹².

In this study, the Climate Data Operators (CDO) were also utilized to remap the datasets, including ERA5, GPM-IMERG, GPM-MERGIR, GSMaP, and NCEP, as well as several forecasts (such as those simulated at 15 km and 60 km horizontal resolutions) to the MONAN forecasts.

¹¹NCAR – National Center for Atmospheric Research, Mesoscale & Microscale Meteorology Laboratory. **Running MPAS-Atmosphere, part 1**. Boulder, CO: NCAR MMM, 2019. 37. [PDF file]. Available at: <https://www2.mmm.ucar.edu/projects/mpas/tutorial/Boulder2019/slides/03a.running_part1.pdf> Accessed on: June 27, 2025

¹²DUDA M. **Model for Ocean-laNd-Atmosphere predictionN (MONAN)**. 2025. Available at: <<https://github.com/mgduda>> Accessed on: June 27, 2025.

4 METHODOLOGY: METRICS AND DIAGNOSTIC FIELDS

Our methodology is designed based on three main characteristics of tropical cyclones commonly utilized in weather forecasting: track, intensity, and rainfall. The metrics that are currently used in similar studies will be described accordingly to each category. Then, for each hurricane, a table with addressed questions will be shown to guide the reader in the analysis and discussion of the results. This table will be part of the workflow, a diagram that shows the steps and figures/maps that will be created. All code made for this dissertation is available on the [GitHub - Dissertation Repository](#), mainly written in Python.

4.1 Trajectory

A map illustrating the forecasted trajectories will be generated using a tracking algorithm developed by the author. A discussion regarding the performance of this tracking and guidance can be found in Appendix ???. Additionally, a time series analysis will be conducted to evaluate the errors associated with each trajectory. These errors are defined as the distance between the central pressure reference points and the forecasted central pressure points, calculated using the GeoPy library ¹. As noted in the literature, the calculations could be performed following the methodology proposed by Moon et al. (2021) but here we simply calculate the great-circle distance between two points.

To quantitatively validate the results, a graphical representation will display the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) for each of the trajectories, regarding the reference dataset.

The MAE is widely employed (DITCHEK et al., 2023; NYONGESA et al., 2024) in the literature and can be defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |F_i - O_i| \quad (4.1)$$

in this equation, N represents the sample size (number of points in the trajectory), F denotes the forecast outputs generated by the model, and O refers to the observation outputs obtained from a reference dataset. The error values can range from 0 to ∞ , with a perfect score indicated by 0. The aim is to calculate the average magnitude

¹GEOPY. **GeoPy's documentation**. 2025. (Online resource) Available at: [<https://geopy.readthedocs.io/en/stable/>](https://geopy.readthedocs.io/en/stable/) Accessed on: June 27, 2025.

of the forecast errors. It is important to note that this error does not convey the direction of the deviations due to its absolute nature; this aspect will be examined using another method discussed later.

Along with the MAE, usually, RMSE is computed to seek a kind of average error, but now weighted according to the square of the error. As the same as before, it varies from 0 to ∞ , and a perfect score means RMSE equal to 0. It is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (F_i - O_i)^2} \quad (4.2)$$

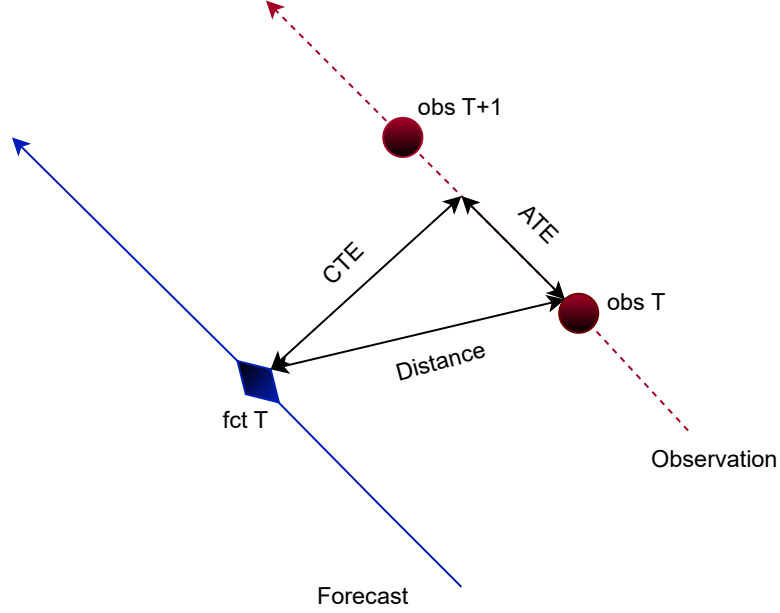
The letters in this equation have the same meaning as in the previous equation. To better understand the potential trends in the trajectory, in addition to visual comparisons, one can utilize cross-track and along-track errors computed into a time series. The errors assess the deviation of the forecasted position from the observed path (cross-track) and the speed of the forecast from the observations (along-track). The cross-track error is measured perpendicularly to the observed trajectory, while the along-track error is measured along the actual course at a specified event. Together, these metrics² can also indicate the directional error of each forecast. Figure 4.1 more clearly illustrates the distinction between cross-track and along-track errors.

In the context of our expected findings, negative (positive) values of cross-track errors indicate that the forecasted center of the hurricane is projected to be located to the west (east), while negative (positive) values of along-track errors suggest that the center is slower (faster) than its intended trajectory.

Such errors can be further investigated through the analysis of meteorological fields, as demonstrated by Gao et al. (2023). For example, examining the 700 hPa geopotential height can help identify large-scale features that contribute to tracking biases. In their study, forecasts exhibiting a pronounced eastward track bias for tropical cyclones were often associated with a notably weaker subtropical high over the North Atlantic. Although this work does not include such an approach due to time constraints, it represents a different approach for future research and is highlighted here for reference.

²**Movable Type Scripts.** 2025. (Online Resource). Available at: <<https://geopy.readthedocs.io/en/stable/>> Accessed on: June 27, 2025.

Figure 4.1 - Illustration of cross-track and along-track errors. The definitions of Cross-Track Error (CTE) and Along-Track Error (ATE) are based on the distance between an observation (obs) and a forecast (fct) at the same valid time (T). This distance is computed as the great-circle distance.



Source: Made by the author (2025).

In conclusion, utilizing a straightforward statistical mean of the errors (specifically the MAE and RMSE) can provide valuable insights into the overall performance of the MONAN simulations. The mean is defined as follows:

$$\text{MEAN} = \frac{1}{E} \sum_{i=1}^E \text{errors}_i \quad (4.3)$$

where E denotes the total number of experiments conducted using the MONAN framework. This approach facilitates a comprehensive understanding of the general behavior exhibited by the MONAN runs and can be compared with ERA5, for instance.

4.2 Intensity

In reviewing the literature, tropical cyclone intensity is defined as the maximum surface wind speed associated with the central pressure (DEMARIA et al., 2007; LANDSEA; FRANKLIN, 2013). However, some studies argue that depending exclusively on wind speed does not fully capture the concept of intensity, suggesting that pressure can also serve as a significant indicator of forecasted intensity (SHEPHERD; WALSH, 2017; HEMING, 2017). This dissertation will adopt the latter approach by evaluating both pressure and wind as intensity indicators.

To conduct these evaluations, we will analyze time series data that includes the central pressure identified by the tracking system and the maximum wind speed, which is derived from the first level of the model. A discussion on how to determine this maximum wind speed in the model can be found in Appendix ???. The time series will serve as a measure of forecast performance, allowing us to assess how closely the computed values align with actual observations and where they occur.

In addition to a visual comparison, we will create a graph containing the MAE and RMSE to quantify the forecasts and rank them based on their effectiveness in illustrating intensity. Additionally, the mean of those errors can also be computed and compared with ERA5 to give us insights into the general MONAN performance.

4.3 Rainfall

The analysis of rainfall for our runs is based on the methodology proposed by Marchok et al. (2007). According to these authors, evaluating forecasted rainfall requires examining three key aspects: the rainfall pattern, the mean and distribution of rain volume, and the extreme values.

To investigate the rainfall pattern, we will utilize the Equitable Threat Score (ETS; (MESINGER, 2008)) along with pattern correlation analysis (namely the Pearson Correlation Coefficient) and visual comparisons using selected snapshots, considering also the related bias of those fields. The mean and distribution of rain volume can be assessed through the computation of the Cumulative Density Function (CDF) and the Probability Density Function (PDF). Extreme rainfall values will be specifically analyzed by examining the 85th percentile computed at the CDF. Lastly, we will discuss a performance score for MONAN in comparison to ERA5 by averaging the results of all experiments. Table 4.1 summarizes the metrics to be computed, followed by a brief explanation of each metric and its intended purpose.

Table 4.1 - Metrics used to evaluate rainfall forecast performance.

Metric	Definition
ETS (Equitable Threat Score)	Measures the skill of categorical forecasts by comparing the number of hits to what would be expected by chance: $\text{ETS} = \frac{H - H_r}{H + F + M - H_r}$ where H is the number of hits, F is false alarms, M is misses, and $H_r = \frac{(H+F)(H+M)}{\text{total}}$ is the expected number of hits due to chance.
Pearson Correlation Coefficient	Measures the spatial similarity between observed and forecasted rainfall fields: $r = \frac{\sum_{i=1}^n (F_i - \bar{F})(O_i - \bar{O})}{\sqrt{\sum_{i=1}^n (F_i - \bar{F})^2 \sum_{i=1}^n (O_i - \bar{O})^2}}$ where F and O are forecasted and observed values, calculated over n data points, where i is the index running from 1 to n .
Bias	Quantifies the difference between forecast and observation: $\text{Bias} = F - O$
CDF (Cumulative Distribution Function)	Represents the probability that a variable takes a value less than or equal to a given threshold: $\text{CDF}(x) = P(X \leq x)$
PDF (Probability Density Function)	Describes the relative likelihood for a variable to take a specific value: $\text{PDF}(x) = \frac{d}{dx} \text{CDF}(x)$

Source: Made by the author (2025).

The ETS evaluates how well forecasted “yes” events correspond to observed “yes” events, while accounting for agreements due to random chance. According to [Marchok et al. \(2007\)](#), this implies that the ETS penalizes a model for overproducing rainfall above a given threshold, even if the rainfall pattern is realistic. For this metric, a hit is defined as an event that was forecasted to occur and did occur; a miss is an event that was not forecasted but occurred; and a false alarm is an event that was forecasted but did not occur.

Pattern correlation is computed as the Pearson correlation coefficient (r) between the forecasted and observed rainfall fields. The correlation ranges from -1 to 1, where: (i) $r = 1$ indicates a perfect positive linear relationship; (ii) $r = -1$, a perfect negative linear relationship; and (iii) $r = 0$ implies no linear relationship between forecast

and observation.

The bias is the difference between the forecast (F) and the observations (O), typically averaged over time or space. A negative bias indicates underestimation, while a positive bias indicates overestimation. The bias can range from negative to positive infinity, with 0 representing a perfect forecast.

Both the CDF and the PDF will be used to analyze rainfall distributions. The CDF shows the cumulative probability of rainfall values up to a certain threshold. For example, the corresponding x-axis value when the y-axis equals to 0.5 (or 50%) on the CDF corresponds to the median rainfall amount. The CDF ranges from 0 to 1 on the y-axis, indicating the proportion of the data below a given threshold on the x-axis. For instance, the 0.85 mark on the CDF corresponds to the 85th percentile, meaning 85% of the data falls below that threshold, and the remaining 15% above it. This is a straightforward way to seek for rainfall extremes.

The PDF, similar to a histogram, describes the probability of observing values within a specific range. It is normalized such that the total area under the curve equals 1. To interpret the PDF, one can compute the area under the curve within a specified range, which represents the probability of a value falling in that interval. Mathematically, the CDF is the integral of the PDF. One should notice that both CDF and PDF can be computed empirically or theoretically. The difference between them is that when computing empirically, one will consider the natural distribution of data, and theoretically, one will consider a mathematical distribution (gamma, Gaussian, etc). In this study, we will utilize the empirical approach.

4.4 Hurricane Beryl workflow

With the hurricane Beryl event, the goal is to investigate the cold pool parameterization alongside sensitivity tests, and with those sensitivity tests, select an optimal parameter configuration for the parameterization.

Table 4.2 outlines the questions designed to guide the examination of each characteristic in alignment with the dissertation's objectives. This approach will help refine the discussion and outputs available in the existing literature.

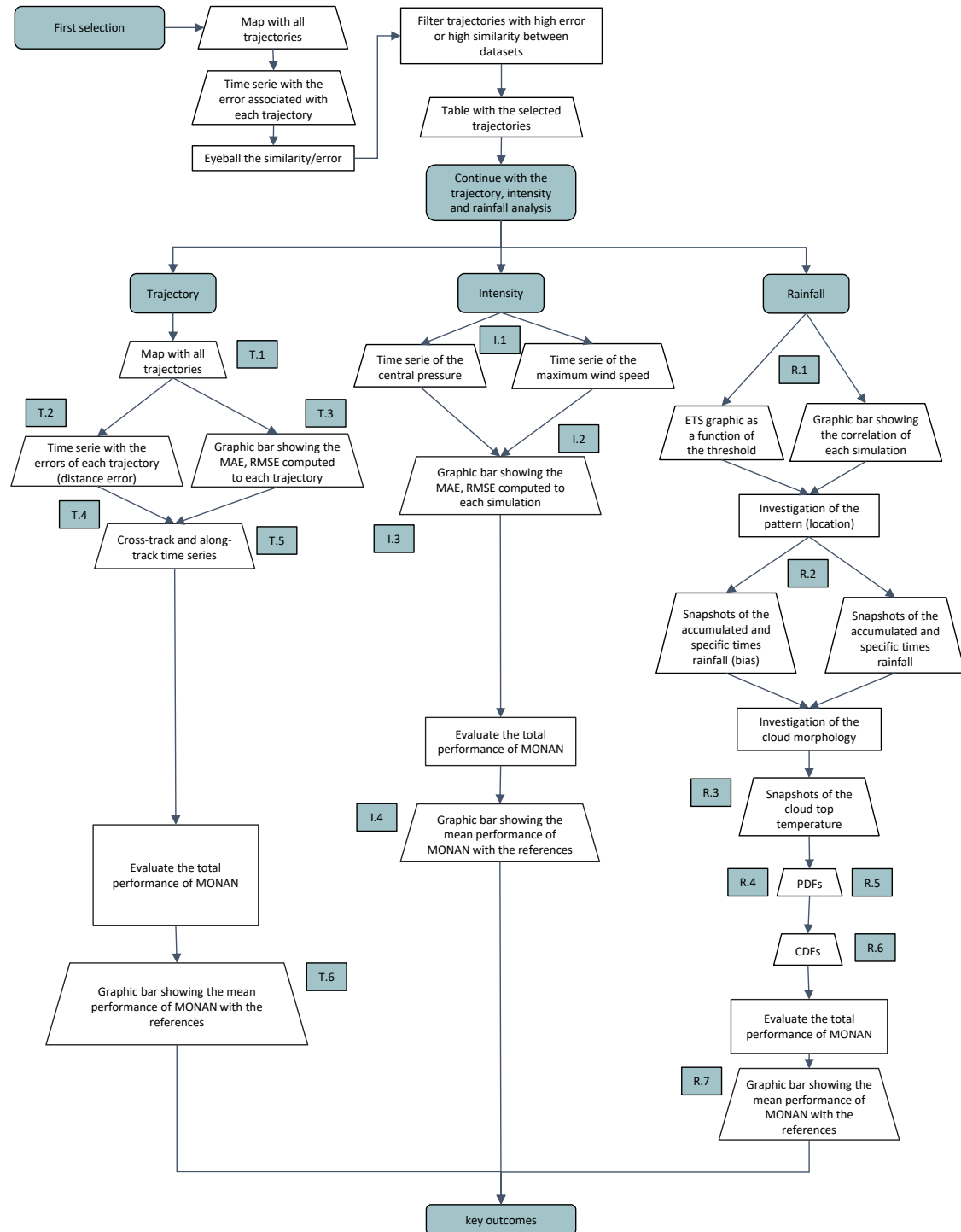
Table 4.2 - Key research questions for track, intensity, and rainfall assessment for Hurricane Beryl.

Topic	Question	ID
Trajectory	What is the overall performance of the parameterization and sensitivity tests in reproducing the hurricane's track?	T.1
	What is the error (in kilometers) associated with these tracks?	T.2
	Which configuration shows the best performance?	T.3
	After how many forecast hours do larger deviations begin to appear?	T.4
	Is there any observable trend in the tracks?	T.5
	What is the overall performance of MONAN?	T.6
Intensity	What is the overall performance of the parameterization and sensitivity tests in reproducing the hurricane's intensity?	I.1
	How much bias is present in these results?	I.2
	How well does ERA5 perform in reproducing this intensity, and why?	I.3
	What is the overall performance of MONAN?	I.4
Rainfall	What is the overall performance of the parameterization and sensitivity tests in reproducing the hurricane's rainfall pattern?	R.1
	In which regions is there a negative (or positive) bias in the rainfall field?	R.2
	What is the overall performance of the parameterization and sensitivity tests in reproducing the hurricane's cloud morphology?	R.3
	How much light, moderate, and heavy rainfall is being produced by the simulations?	R.4
	What is the average rainfall produced by the simulations?	R.5
	What is the bias between the extreme rainfall produced by the simulations and the reference data?	R.6
	What is the overall performance of MONAN?	R.7

Source: Made by the author (2025).

The workflow below summarizes and organizes the upcoming steps. After a quick initial analysis, the results will be visually represented as trapezoidal shapes, while discussions will be depicted with rectangular shapes. This distinction helps clarify the types of information being presented at each stage of the process. Following this, the boxes containing question IDs, as outlined in Table 4.2, related to the three aspects under evaluation (trajectory, intensity, and rainfall) will also be displayed in the workflow displayed in Figure 4.2.

Figure 4.2 - Workflow overview of Hurricane Berryl.



Source: Made by the author (2025).

The reader should note that an initial selection of the results will be conducted. As the diagram indicates, a trajectory map will be created, accompanied by distance errors. The selection criteria will focus on experiments that do not contribute significantly to the discussion, either because they are part of a set of similar experiments or due to excessively large errors that do not accurately represent the model state we aim to address.

4.5 Hurricane Helene workflow

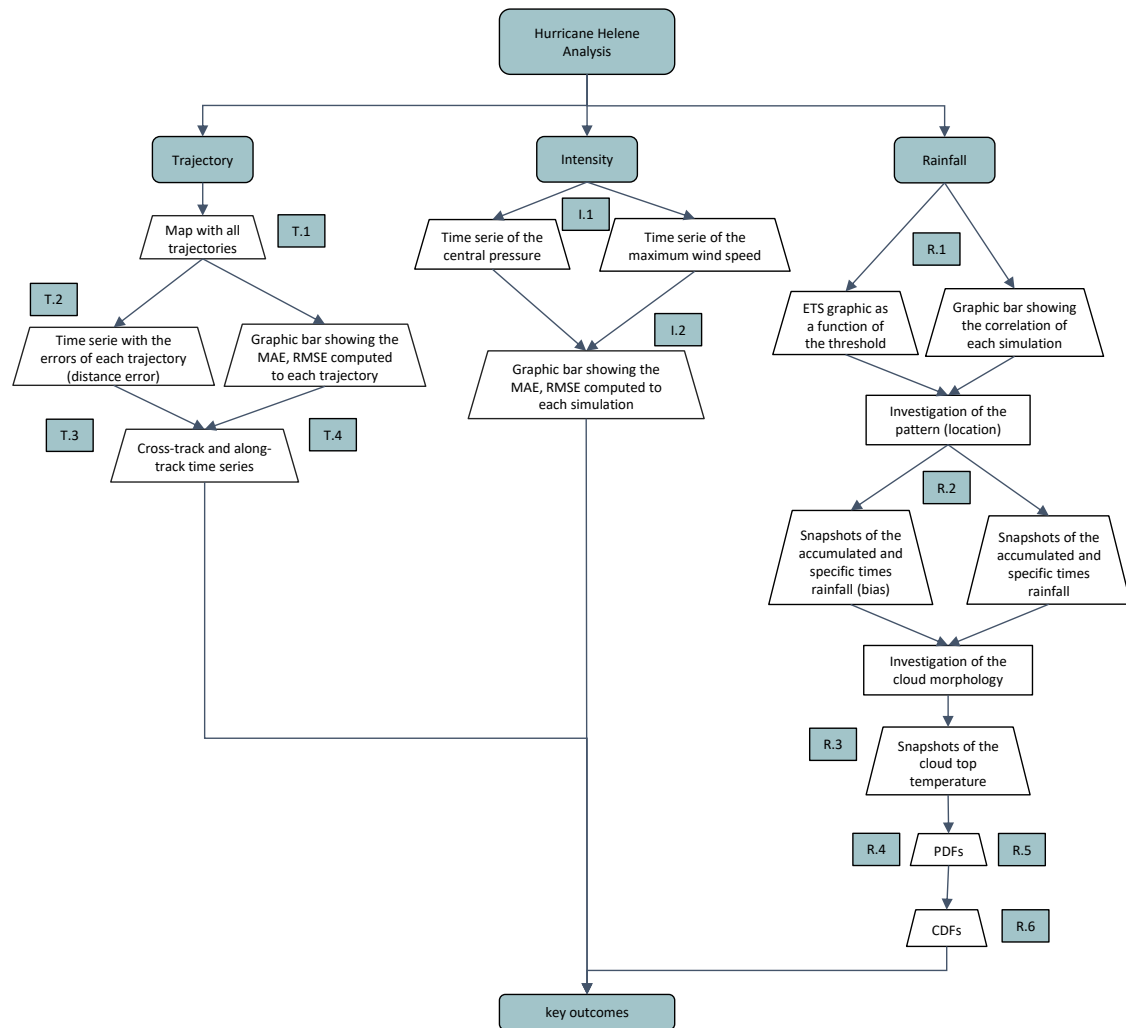
Now we aim to test the best configuration alongside the cold pool effect and the highest resolution effect in another hurricane, Helene. Table 4.3 indicates the questions addressed for this investigation, and furthermore, a similar diagram as designed in the previous section were made here in Figure 4.3.

Table 4.3 - Key research questions for track, intensity, and rainfall assessment for Hurricane Helene.

Topic	Question	ID
Trajectory	What is the overall performance of the optimal configuration found, the parameterization, and the high-resolution effect in reproducing the hurricane's track?	T.1
	What is the error (in kilometers) associated with these tracks?	T.2
	After how many forecast hours do larger deviations begin to appear?	T.3
	Is there any observable trend in the tracks?	T.4
Intensity	What is the overall performance of the optimal configuration found, the parameterization, and the high-resolution effect in reproducing the hurricane's intensity?	I.1
	How much bias is present in these results?	I.2
Rainfall	What is the overall performance of the optimal configuration found, the parameterization, and the high-resolution effect in reproducing the hurricane's rainfall pattern?	R.1
	In which regions is there a negative (or positive) bias in the rainfall field?	R.2
	What is the overall performance of the optimal configuration found, the parameterization, and the high-resolution effect in reproducing the hurricane's cloud morphology?	R.3
	How much light, moderate, and heavy rainfall is being produced by the simulations?	R.4
	What is the average rainfall produced by the simulations?	R.5
	What is the bias between the extreme rainfall produced by the simulations and the reference data? Do the forecasts capture the extreme event in North Carolina?	R.6

Source: Made by the author (2025).

Figure 4.3 - Workflow overview of Hurricane Helene.



Source: Made by the author (2025).

REFERENCES

- ARAKAWA, A. The cumulus parameterization problem: Past, present, and future. **Journal of climate**, v. 17, n. 13, p. 2493–2525, 2004. 1
- ARAKAWA, A.; JUNG, J.-H.; WU, C.-M. Toward unification of the multiscale modeling of the atmosphere. **Atmospheric Chemistry and Physics**, Copernicus Publications Göttingen, Germany, v. 11, n. 8, p. 3731–3742, 2011. 1
- ARAKAWA, A.; SCHUBERT, W. H. Interaction of a cumulus cloud ensemble with the large-scale environment, part i. **Journal of Atmospheric Sciences**, v. 31, n. 3, p. 674–701, 1974. 1, 18, 21
- BAGTASA, G. Assessment of tropical cyclone rainfall from gsmap and gpm products and their application to analog forecasting in the philippines. **Atmosphere**, MDPI, v. 13, n. 9, p. 1398, 2022. 37
- BARNES, G.; ZIPSER, E.; JORGENSEN, D.; JR, F. M. Mesoscale and convective structure of a hurricane rainband. **Journal of Atmospheric Sciences**, v. 40, n. 9, p. 2125–2137, 1983. 27
- BECHTOLD, P.; SEMANE, N.; LOPEZ, P.; CHABOUREAU, J.-P.; BELJAARS, A.; BORMANN, N. Representing equilibrium and nonequilibrium convection in large-scale models. **Journal of the Atmospheric Sciences**, v. 71, n. 2, p. 734–753, 2014. 20
- BOPAPE, M.-J. M.; CARDOSO, H.; PLANT, R. S.; PHADULI, E.; CHIKOORE, H.; NDARANA, T.; KHALAU, L.; RAKATE, E. Sensitivity of tropical cyclone idai simulations to cumulus parametrization schemes. **Atmosphere**, MDPI, v. 12, n. 8, p. 932, 2021. 33
- BRACKINS, J. T.; KALYANAPU, A. J. Evaluation of parametric precipitation models in reproducing tropical cyclone rainfall patterns. **Journal of Hydrology**, Elsevier, v. 580, p. 124255, 2020.
- CAI, Q.; TANG, X. Effect of the eyewall cold pool on the inner rainband of a tropical cyclone. **Journal of Geophysical Research: Atmospheres**, Wiley Online Library, v. 124, n. 3, p. 1292–1306, 2019. 2, 28, 31
- CENTER, F. S. U. C. **Hurricane Helene Post-Storm Summary Report**. 2024. Accessed: 2025-05-24. Available from: <<<https://climatecenter.fsu.edu/images/docs/Hurricane-Helene-Summary-Report.pdf>>>.
- CHABOUREAU, J.-P.; BECHTOLD, P. Statistical representation of clouds in a regional model and the impact on the diurnal cycle of convection during tropical convection, cirrus and nitrogen oxides (troccinox). **Journal of Geophysical Research: Atmospheres**, v. 110, n. D17, 2005. Available from: <<<https://doi.org/10.1029/2004JD005645>>>. 40

- CHEN, B.-F.; ELSBERRY, R. L.; LEE, C.-S. Origin and maintenance of the long-lasting, outer mesoscale convective system in typhoon fengshen (2008). **Monthly Weather Review**, v. 142, n. 8, p. 2838–2859, 2014. [2](#), [29](#), [31](#)
- DEMARIA, M.; KNAFF, J.; SAMPSON, C. Evaluation of long-term trends in tropical cyclone intensity forecasts. **Meteorology and Atmospheric Physics**, Springer, v. 97, n. 1, p. 19–28, 2007. [48](#)
- DITCHEK, S. D.; SIPPEL, J. A.; MARINESCU, P. J.; JR, G. J. A. Improving best track verification of tropical cyclones: A new metric to identify forecast consistency. **Weather and Forecasting**, v. 38, n. 6, p. 817–831, 2023. [45](#)
- DROEGEMEIER, K. K.; WILHELMSON, R. B. Numerical simulation of thunderstorm outflow dynamics. part i: Outflow sensitivity experiments and turbulence dynamics. **Journal of atmospheric sciences**, v. 44, n. 8, p. 1180–1210, 1987. [20](#)
- DULAC, W.; CATTIAUX, J.; CHAUVIN, F.; BOURDIN, S.; FROMANG, S. Assessing the representation of tropical cyclones in era5 with the cnrm tracker. **Climate Dynamics**, Springer, v. 62, n. 1, p. 223–238, 2024. [33](#), [34](#)
- EASTIN, M. D.; GARDNER, T. L.; LINK, M. C.; SMITH, K. C. Surface cold pools in the outer rainbands of tropical storm hanna (2008) near landfall. **Monthly Weather Review**, v. 140, n. 2, p. 471–491, 2012. [27](#)
- EMANUEL, K. A. **Atmospheric convection**. [S.l.]: Oxford university press, 1994. [8](#), [20](#)
- FENG, Z.; HAGOS, S.; ROWE, A. K.; BURLEYSON, C. D.; MARTINI, M. N.; SZOEKE, S. P. de. Mechanisms of convective cloud organization by cold pools over tropical warm ocean during the amie/dynamo field campaign. **Journal of Advances in Modeling Earth Systems**, Wiley Online Library, v. 7, n. 2, p. 357–381, 2015. [1](#), [10](#), [11](#), [12](#), [13](#), [30](#)
- FREITAS, S. R.; GRELL, G. A.; CHOVERT, A. D.; DIAS, M. A. F. S.; NASCIMENTO, E. de L. A parameterization for cloud organization and propagation by evaporation-driven cold pool edges. **Journal of Advances in Modeling Earth Systems**, Wiley Online Library, v. 16, n. 1, p. e2023MS003982, 2024. [1](#), [2](#), [3](#), [13](#), [16](#), [17](#), [18](#), [20](#), [22](#), [30](#), [39](#), [40](#)
- FREITAS, S. R.; GRELL, G. A.; LI, H. The gf convection parameterization: Recent developments, extensions, and applications. **Geoscientific Model Development Discussions**, Göttingen, Germany, v. 2020, p. 1–28, 2020. [21](#)
- FREITAS, S. R.; GRELL, G. A.; MOLOD, A.; THOMPSON, M. A.; PUTMAN, W. M.; SILVA, C. M. Santos e; SOUZA, E. P. Assessing the grell-freitas convection parameterization in the nasa geos modeling system. **Journal of Advances in Modeling Earth Systems**, Wiley Online Library, v. 10, n. 6, p. 1266–1289, 2018. [20](#), [40](#)

FREITAS, S. R.; PANETTA, J.; LONGO, K. M.; RODRIGUES, L. F.; MOREIRA, D. S.; ROSARIO, N. E.; DIAS, P. L. S.; DIAS, M. A. S.; SOUZA, E. P.; FREITAS, E. D. et al. The brazilian developments on the regional atmospheric modeling system (brams 5.2): an integrated environmental model tuned for tropical areas. **Geoscientific Model Development**, Copernicus Publications Göttingen, Germany, v. 10, n. 1, p. 189–222, 2017. 16

FUGLESTVEDT, H. F.; HAERTER, J. O. Cold pools as conveyor belts of moisture. **Geophysical Research Letters**, Wiley Online Library, v. 47, n. 12, p. e2020GL087319, 2020. 10

GAO, K.; HARRIS, L.; BENDER, M.; CHEN, J.-H.; ZHOU, L.; KNUTSON, T. Regulating fine-scale resolved convection in high-resolution models for better hurricane track prediction. **Geophysical Research Letters**, Wiley Online Library, v. 50, n. 13, p. e2023GL103329, 2023. 46

GENIO, A. D. D.; CHEN, Y.; KIM, D.; YAO, M.-S. Constraints on cumulus parameterization from simulations of observed mjo events. **Journal of Climate**, v. 28, n. 16, p. 6419–6442, 2015. 14, 30

GRANDPEIX, J.-Y.; LAFORE, J.-P. A density current parameterization coupled with emanuel’s convection scheme. part i: The models. **Journal of the Atmospheric Sciences**, v. 67, n. 4, p. 881–897, 2010. 13, 30

GRANDPEIX, J.-Y.; LAFORE, J.-P.; CHERUY, F. A density current parameterization coupled with emanuel’s convection scheme. part ii: 1d simulations. **Journal of the Atmospheric Sciences**, v. 67, n. 4, p. 898–922, 2010. 13

GRANT, L. D.; HEEVER, S. C. van den. Cold pool-land surface interactions in a dry continental environment. **Journal of Advances in Modeling Earth Systems**, Wiley Online Library, v. 10, n. 7, p. 1513–1526, 2018. 2

GRANT, L. D.; LANE, T. P.; HEEVER, S. C. van den. The role of cold pools in tropical oceanic convective systems. **Journal of the Atmospheric Sciences**, v. 75, n. 8, p. 2615–2634, 2018. 28, 31

GRELL, G. A.; DÉVÉNYI, D. A generalized approach to parameterizing convection combining ensemble and data assimilation techniques. **Geophysical Research Letters**, Wiley Online Library, v. 29, n. 14, p. 38–1, 2002. 21

GRELL, G. A.; FREITAS, S. R. A scale and aerosol aware stochastic convective parameterization for weather and air quality modeling. **Atmospheric Chemistry and Physics**, Copernicus GmbH, v. 14, n. 10, p. 5233–5250, 2014. 1, 20, 21, 40

HAGEN, A. B.; , C. ; CANGIALOSI, J. P.; CHENARD, M.; ALAKA, L.; DELGADO, S. **Tropical Cyclone Report: Hurricane Helene (AL092024), 24 – 27 September 2024**. [S.l.], april 2025. Accessed: 2025-06-24. Available from: <<https://www.nhc.noaa.gov/data/tcr/AL092024_Helene.pdf>>. 65

- HEMING, J. Tropical cyclone tracking and verification techniques for met office numerical weather prediction models. **Meteorological Applications**, Wiley Online Library, v. 24, n. 1, p. 1–8, 2017. 48
- HIRT, M.; CRAIG, G. C. A cold pool perturbation scheme to improve convective initiation in convection-permitting models. **Quarterly Journal of the Royal Meteorological Society**, Wiley Online Library, v. 147, n. 737, p. 2429–2447, 2021. 14
- HONG, S.-Y. A new stable boundary-layer mixing scheme and its impact on the simulated east asian summer monsoon. **Quarterly Journal of the Royal Meteorological Society**, v. 136, n. 651, p. 1481–1496, 2010. Available from: <<<https://doi.org/10.1002/qj.665>>>. 40
- HONG, S.-Y.; LIM, J.-O. J. The wrf single-moment 6-class microphysics scheme (wsm6). **Asia-Pacific Journal of Atmospheric Sciences**, v. 42, n. 2, p. 129–151, 2006. 40
- Houze Jr, R. A. Clouds in tropical cyclones. **Monthly Weather Review**, v. 138, n. 2, p. 293–344, 2010. 24, 26
- HUFFMAN, G. Gpm imerg final precipitation l3 half hourly 0.1 degree× 0.1 degree v06. (No Title), 2019. 35
- IACONO, M. J.; DELAMERE, J. S.; MLAWER, E. J.; SHEPHARD, M. W.; CLOUGH, S. A.; COLLINS, W. D. Radiative forcing by long-lived greenhouse gases: Calculations with the aer radiative transfer models. **Journal of Geophysical Research: Atmospheres**, Wiley Online Library, v. 113, n. D13, 2008. 40
- JANOWIAK, J.; JOYCE, B.; XIE, P. **NCEP/CPC L3 half hourly 4 km global (60S–60N) merged IR V1**. Goddard Earth Sciences Data and Information Services Center (GES DISC). 2017. 36
- KESSLER, E. **Thunderstorm morphology and dynamics**. [S.l.]: US Department of Commerce, National Oceanic and Atmospheric Administration . . . , 1982. 8
- KNAPP, K. R.; KRUK, M. C.; LEVINSON, D. H.; DIAMOND, H. J.; NEUMANN, C. J. The international best track archive for climate stewardship (ibtracs) unifying tropical cyclone data. **Bulletin of the American Meteorological Society**, American Meteorological Society, v. 91, n. 3, p. 363–376, 2010. 33
- LANDSEA, C. W.; FRANKLIN, J. L. Atlantic hurricane database uncertainty and presentation of a new database format. **Monthly Weather Review**, v. 141, n. 10, p. 3576–3592, 2013. 48
- LANGHANS, W.; ROMPS, D. M. The origin of water vapor rings in tropical oceanic cold pools. **Geophysical Research Letters**, Wiley Online Library, v. 42, n. 18, p. 7825–7834, 2015. 10, 11

- LIN, J.; QIAN, T.; KLOTZBACH, P. Tropical cyclones. **Atmosphere-Ocean**, Taylor & Francis, v. 60, n. 3-4, p. 360–398, 2022. 24, 26, 27
- MALLINSON, H. M.; LASHER-TRAPP, S. G. An investigation of hydrometeor latent cooling upon convective cold pool formation, sustainment, and properties. **Monthly Weather Review**, v. 147, n. 9, p. 3205–3222, 2019. 7
- MARCHOK, T.; ROGERS, R.; TULEYA, R. Validation schemes for tropical cyclone quantitative precipitation forecasts: Evaluation of operational models for us landfalling cases. **Weather and forecasting**, v. 22, n. 4, p. 726–746, 2007. 48, 49
- MAY, J.; HARANDI, M.; TYO, J. S.; HU, L.; RITCHIE-TYO, E. A. A cnn system for segmenting tropical cyclones neighborhoods in geostationary images. In: IEEE. **IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium**. [S.l.], 2024. p. 8259–8262. 33
- MAZZA, E.; CHEN, S. S. Tropical cyclone rainfall climatology, extremes, and flooding potential from remote sensing and reanalysis datasets over the continental united states. **Journal of Hydrometeorology**, American Meteorological Society, v. 24, n. 9, p. 1549–1562, 2023. 33, 38
- MESINGER, F. Bias adjusted precipitation threat scores. **Advances in Geosciences**, Copernicus Publications Göttingen, Germany, v. 16, p. 137–142, 2008. 48
- MITCHELL, K. The community noah land-surface model (lsm). **User’s Guide public release version**, v. 2, n. 1, p. 3660–3667, 2005. 40
- MOON, J.; PARK, J.; CHA, D.-H.; MOON, Y. Five-day track forecast skills of wrf model for the western north pacific tropical cyclones. **Weather and Forecasting**, v. 36, n. 4, p. 1491–1503, 2021. 45
- MULLER, C.; TAKAYABU, Y. Response of precipitation extremes to warming: what have we learned from theory and idealized cloud-resolving simulations, and what remains to be learned? **Environmental Research Letters**, IOP Publishing, v. 15, n. 3, p. 035001, 2020. 5, 7
- NAKANISHI, M.; NIINO, H. Development of an improved turbulence closure model for the atmospheric boundary layer. **Journal of the Meteorological Society of Japan. Ser. II**, Meteorological Society of Japan, v. 87, n. 5, p. 895–912, 2009. 40
- National Hurricane Center. **Tropical Cyclone Report: Hurricane Helene (AL092024)**. [S.l.], 2024. Acesso em: 24 maio 2025. Available from: <<https://www.nhc.noaa.gov/data/tcr/AL092024_Helene.pdf>>.
- National Hurricane Center (NHC) and Central Pacific Hurricane Center. **Saffir–Simpson Hurricane Wind Scale**. 2025. <<https://www.nhc.noaa.gov/aboutsshws.php>>. Accessed on: June 27, 2025. 25

National Oceanic and Atmospheric Administration (NOAA). **2024: An active year of U.S. billion-dollar weather and climate disasters**. January 2025. <<https://www.climate.gov/news-features/blogs/beyond-data/2024-active-year-us-billion-dollar-weather-and-climate-disasters>>. Accessed on: July 27, 2025. 2

_____. **Hurricane Costs**. 2025. <<https://coast.noaa.gov/states/fast-facts/hurricane-costs.html>>. Accessed on: July 27, 2025. 2

_____. **Tropical Cyclone Introduction**. 2025. <<https://www.noaa.gov/jetstream/tropical/tropical-cyclone-introduction>>. Accessed: 2025-06-24. 25

NOAA EMC. **Parallel HAFS Ensemble Forecast Guidance for Current Active Storms**. 2025. <<https://www.emc.ncep.noaa.gov/HAFS/HAFSEPS/index.php>>. Accessed: 2025-06-27.

NOAA STAR. **Center for Satellite Applications and Research (STAR)**. 2025. <<https://www.star.nesdis.noaa.gov/star/index.php>>. Accessed: 2025-06-27.

NSF NCAR/MMM. **Plotting MPAS-A Global Simulations**. Boulder, CO: [s.n.], 2024. Available at: <<https://www2.mmm.ucar.edu/projects/mpas/site/visualization.html>>. Accessed on: June 27, 2025. 41

NYONGESA, A. M.; SHI, D.; LI, S.; LI, Q. Influence of convective parameterization on the simulation of tropical cyclones over the south west indian ocean: A case study of tropical cyclone idai (2019). **Atmospheric Research**, Elsevier, v. 306, p. 107461, 2024. 45

PÉREZ-ALARCÓN, A.; FERNÁNDEZ-ALVAREZ, J. C.; BATISTA-LEYVA, A. J. Performance evaluation of numerical tools for hurricane forecast (nthf) system during 2020 north atlantic tropical cyclones season. **Environmental Sciences Proceedings**, MDPI, v. 8, n. 1, p. 22, 2021. 24, 26

RAFFERTY, E. J. P. **Hurricane Helene**. 2025. <<https://www.britannica.com/event/Hurricane-Helene>>. Accessed: 2025-05-24.

REDDY, M. V.; MITRA, A. K.; MOMIN, I.; KRISHNA, U. M. How accurately satellite precipitation products capture the tropical cyclone rainfall? **Journal of the Indian Society of Remote Sensing**, Springer, v. 50, n. 10, p. 1871–1884, 2022. 37

ROONEY, G. G.; STIRLING, A. J.; STRATTON, R. A.; WHITALL, M. C-pool: A scheme for modelling convective cold pools in the met office unified model. **Quarterly Journal of the Royal Meteorological Society**, Wiley Online Library, v. 148, n. 743, p. 962–980, 2022. 15

- SAFFIR, H. S. Hurricane wind and storm surge. **The Military Engineer**, JSTOR, v. 65, n. 423, p. 4–5, 1973. 24
- SAWADA, M.; IWASAKI, T. Impacts of evaporation from raindrops on tropical cyclones. part ii: Features of rainbands and asymmetric structure. **Journal of the atmospheric sciences**, v. 67, n. 1, p. 84–96, 2010. 2, 27, 31
- _____. _____. **Journal of the atmospheric sciences**, v. 67, n. 1, p. 84–96, 2010. 27, 28
- SHEPHERD, T. J.; WALSH, K. J. Sensitivity of hurricane track to cumulus parameterization schemes in the wrf model for three intense tropical cyclones: Impact of convective asymmetry. **Meteorology and Atmospheric Physics**, Springer, v. 129, p. 345–374, 2017. 48
- SIMPSON, R. H. The hurricane disaster—potential scale. **Weatherwise**, Informa UK Limited, v. 27, n. 4, p. 169–186, 1974. 24
- SKAMAROCK, W. C.; KLEMP, J. B.; DUDA, M. G.; FOWLER, L. D.; PARK, S.-H.; RINGLER, T. D. A multiscale nonhydrostatic atmospheric model using centroidal voronoi tessellations and c-grid staggering. **Monthly Weather Review**, v. 140, n. 9, p. 3090–3105, 2012. 2, 41
- STENSRUD, D. J. **Parameterization schemes: keys to understanding numerical weather prediction models**. [S.l.]: Cambridge University Press, 2007. 1
- TOMPKINS, A. M. Organization of tropical convection in low vertical wind shears: The role of water vapor. **Journal of the atmospheric sciences**, v. 58, n. 6, p. 529–545, 2001. 1, 8, 9, 10, 11, 30
- TORRI, G.; KUANG, Z.; TIAN, Y. Mechanisms for convection triggering by cold pools. **Geophysical Research Letters**, Wiley Online Library, v. 42, n. 6, p. 1943–1950, 2015. 1, 10, 11, 12, 30
- UK Met Office. **Observed and forecast tracks: northern hemisphere 2024**. 2024. <<https://www.metoffice.gov.uk/research/weather/tropical-cyclones/verification/seasons/nhem2024>>. Accessed: 2025-06-27.
- WEISMAN, M. L.; ROTUNNO, R. “a theory for strong long-lived squall lines” revisited. **Journal of the Atmospheric Sciences**, v. 61, n. 4, p. 361–382, 2004. 8, 11
- World Meteorological Organization (WMO). **Classification of tropical cyclones**. sep. 2023. <<https://wmo.int/content/classification-of-tropical-cyclones>>. Accessed on: July 27, 2025. 24
- YAMAGUCHI, M.; NAKAZAWA, T.; AONASHI, K. Tropical cyclone track forecasts using jma model with ecmwf and jma initial conditions. **Geophysical research letters**, Wiley Online Library, v. 39, n. 9, 2012.

YANG, S.; SURRATT, M.; WHITCOMB, T. R.; CAMACHO, C. Evaluation of imerg and gsmep for tropical cyclone applications. **Geophysical Research Letters**, Wiley Online Library, v. 51, n. 4, p. e2023GL106414, 2024. 33, 37

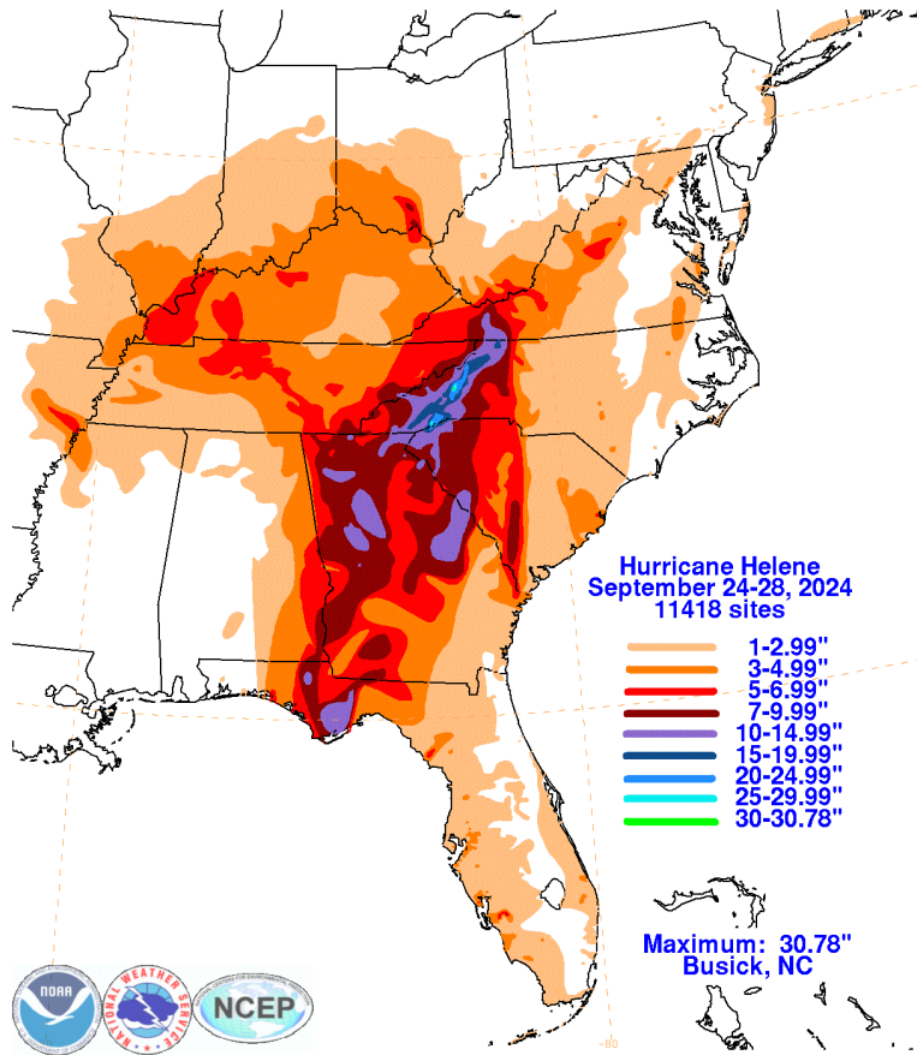
ZEHNDER, J. A. **Tropical Cyclone**. Chicago, IL: [s.n.], 2025. Fact-checked by the editors. <<https://www.britannica.com/science/tropical-cyclone>>. Accessed on: June 27, 2025. 26

ZHOU, Y.; MATYAS, C. J. Regionalization of precipitation associated with tropical cyclones using spatial metrics and satellite precipitation. **GIScience & Remote Sensing**, Taylor & Francis, v. 58, n. 4, p. 542–561, 2021. 33

APPENDIX A - OVERVIEW OF THE DATASETS IN REPRODUCING EXTREMES

Figure A.1 was taken from the Hurricane Helene report (HAGEN et al., 2025). It shows the accumulation of rainfall from September 24th to September 28th.

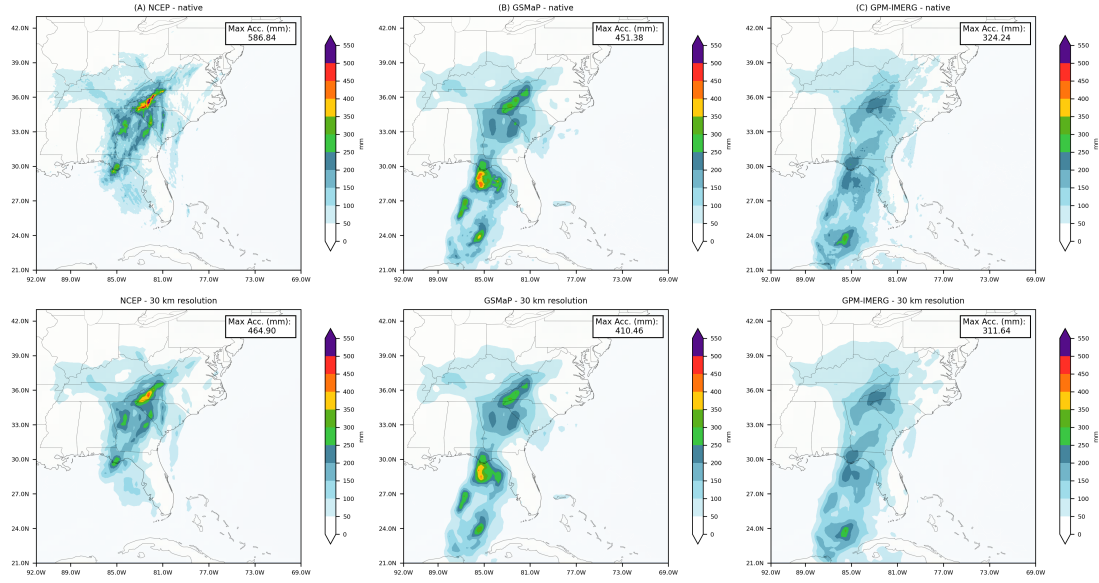
Figure A.1 - Analysis of storm total rainfall (inches) for Hurricane Helene.



Source: Hagen et al. (2025).

Keeping this in mind, we tried to reproduce this figure with NCEP, GPM-IMERG, and GSMaP. Figure A.2 shows the panels at their native resolution in the first line, and with the model resolution (30 km) in the second line.

Figure A.2 - Panels of the accumulated rainfall from September 25th (12 UTC) until September 29th (00 UTC).



Source: Made by the author (2025).

By comparing Figures A.1 with A.2, one could notice that the dataset coming from NCEP is the most similar to Figure A.1, followed by GSMaP. It is not mentioned where Figure A.1 data was taken from, but we hypothesized that it comes from the same source, since NCEP encompasses the stations and radar datasets. GPM-IMERG, however, fails to represent the extremes shown in both NCEP and GSMaP, and can't capture some structures of the rain (85-81°W and 32-27°N) that NCEP and GSMaP can. For our analysis, we used NCEP data over the United States and GSMaP outside the United States, which seems to better represent the amount produced and some structures as well. One should keep in mind that the downscaling spreads the data, creating a systematic bias (panels A and C).

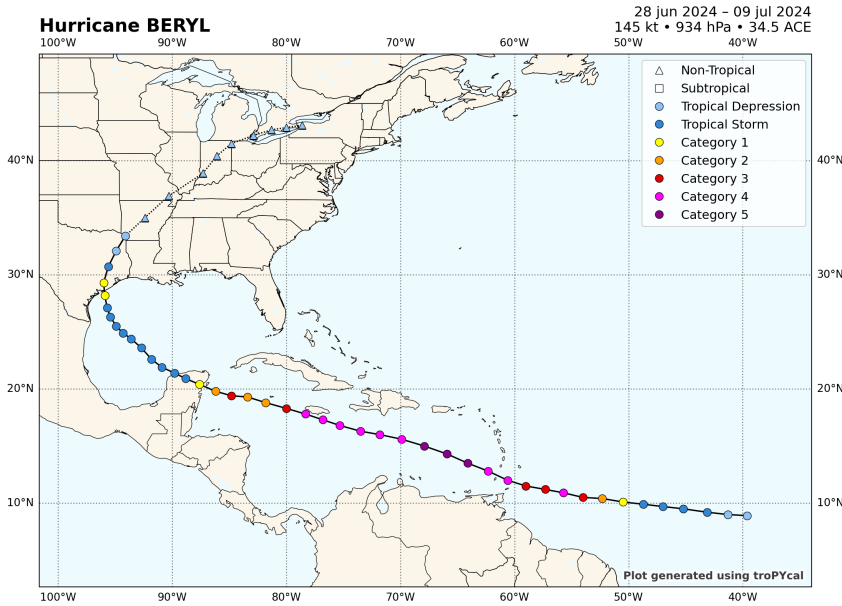
APPENDIX B - TRACKING ALGORITHM

In this appendix, we will discuss how the algorithm for tracking was conceived. All codes concerning this dissertation are available on the author’s Github: <<https://github.com/biancafusi/Dissertation>>. A brief explanation of the steps of how this code works will be provided below.

Inicialmente partimos dos dados do IBTrACS. A Figura B.1, mostra os pontos de minimo Mean Sea Level Pressure (MSLP) plotados durante todo o período do Beryl.

Initially, we gathered the data from IBTrACS. Figure B.1 shows the points plotted during the whole period that Hurricane Beryl was active, where the MLSP is minimal.

Figure B.1 - Tracking of Hurricane Beryl.



Source: Made by the author (2025).

As for the code, the initial step is to store the coordinates (latitude and longitude). For that purpose, the variables `lat_points_NOAA` and `lon_points_NOAA` were created, studying the period from July 3rd to July 9th. Following, there will be slices of the original code, concerning each step explained.

```
# Reference dataset: best track (NOAA)
```

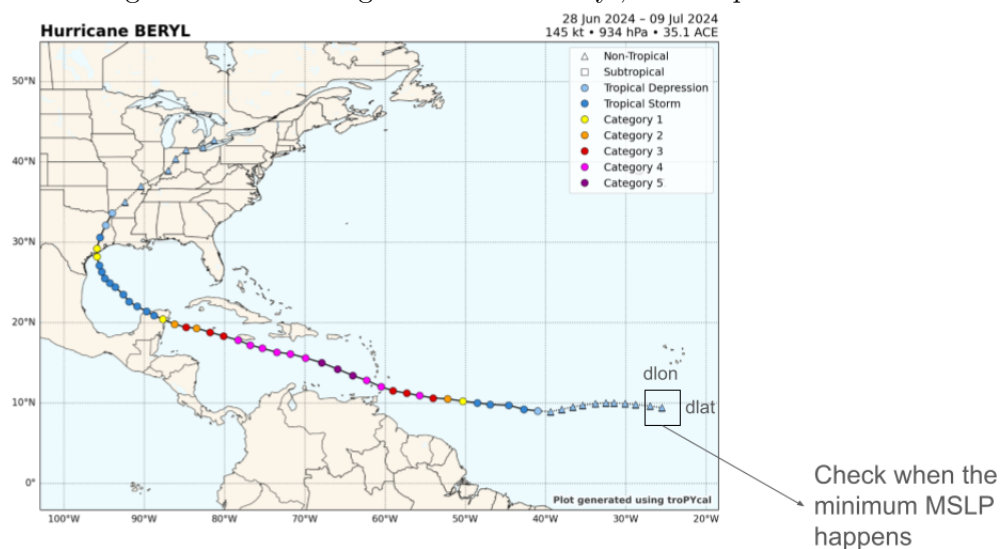
```

2 NOAA = xr.open_dataset(NOAA_path)
3 time = NOAA.time.sel(time=slice(initial_day, final_day))
4 NOAA_hourly = NOAA.sel(time=time)
5 MSLP = NOAA_hourly.mslp
6 lat = NOAA_hourly.lat
7 lon = NOAA_hourly.lon
8 # Stores NOAA's tracking
9 lat_points_NOAA, lon_points_NOAA, mslp_points_NOAA = [], [],
  []
10 for t in range(0, len(time), 1):
11     lon_array_sel = lon.isel(time=t)
12     lat_array_sel = lat.isel(time=t)
13     lon_points_NOAA.append(lon_array_sel.values)
14     lat_points_NOAA.append(lat_array_sel.values)
15     mslp_points_NOAA.append(MSLP.isel(time=t).values)

```

Having these values, we can draw an initial box around these points. Figure B.2 shows how these boxes appear on the tracking.

Figure B.2 - Tracking of Hurricane Beryl, first step of the code.



Source: Made by the author (2025).

The following slice of code describes the process to draw these boxes:

```

1 dlat, dlon = 2.0, 2.0

```

```

2
3 # [...] more code before
4 for t in range(len(time)):
5     MSLP_t = dataset_MSLP.isel(time=t)
6     WSPD_t = dataset_WSPD.isel(time=t)
7
8     if label == 'NOAA':
9         lon_array_sel = lon_array.isel(time=t)
10        lat_array_sel = lat_array.isel(time=t)
11
12        lon_points.append(lon_array_sel.values)
13        lat_points.append(lat_array_sel.values)
14
15        mslp_points.append(MSLP_t.values)
16        wspd_points.append(WSPD_t.values)
17        time_steps.append(t)
18
19        lat_points_NOAA = lat_points
20        lon_points_NOAA = lon_points
21
22    else:
23        # Searching for the minimal value (based on NOAA)
24        upper_lat, lower_lat = lat_points_NOAA[t] + dlat,
25                               lat_points_NOAA[t] - dlat
26        left_lon, right_lon = lon_points_NOAA[t] - dlon,
27                               lon_points_NOAA[t] + dlon
28
29        lon_sliced = lon_array.sel(lon=slice(left_lon,
30                                              right_lon))
31        if label == 'ERA5':
32            lat_sliced = lat_array.sel(lat=slice(
33                upper_lat, lower_lat))
34            MSLP_sliced = MSLP_t.sel(lon=slice(left_lon,
35                                              right_lon), lat=slice(upper_lat, lower_lat)
36                                   )
37        else:
38            lat_sliced = lat_array.sel(lat=slice(
39                lower_lat, upper_lat))
40            MSLP_sliced = MSLP_t.sel(lon=slice(left_lon,

```

```

35         right_lon), lat=slice(lower_lat, upper_lat)
36     )
37     lat_t, lon_t = np.meshgrid(lat_sliced, lon_sliced
        , indexing='ij')

```

One should notice the `upper_lat`; `lower_lat` and `left_lon`; `right_lon`, which are the limits of these box, are made using the information from `dlat` and `dlon`.

The next slice of code searches inside these boxes, looking for the minimal value of MSLP from the referred dataset, for instance, ERA5 or MONAN outputs:

```

1  # [...] more code before
2
3      # Verify existing points within the selected
      region
4      if MSLP_sliced.size > 0 and not np.isnan(
        MSLP_sliced).all():
5
6          # Finds the index relative to minimum values
          within th ebox
7          min_index = np.nanargmin(MSLP_sliced.values)
          # ignores nan
8          min_value = MSLP_sliced.values.ravel()[
            min_index]    # minimum value
9
10         # Converts from 1D two 2D
11         lat_index, lon_index = np.unravel_index(
            min_index, MSLP_sliced.values.shape)
12
13         # Gets lat/lon points relative to each
            minimum value
14         lat_sel = lat_t[lat_index, lon_index]
15         lon_sel = lon_t[lat_index, lon_index]
16
17         # Add those values in lists
18         mslp_points.append(min_value)
19         lat_points.append(lat_sel)
20         lon_points.append(lon_sel)

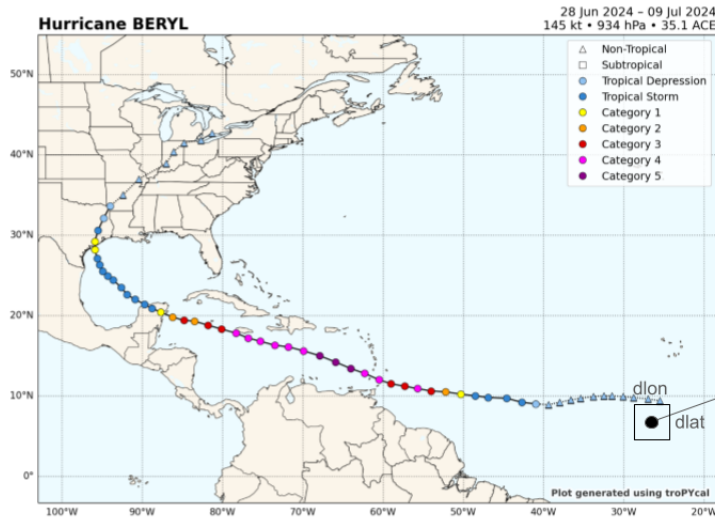
```



```
time_steps.append(t)
```

Once this minimal value is found, we create a second box around this value in order to find the maximum wind speed. Figure B.3 shows how these process appears in the tracking.

Figure B.3 - Tracking of Hurricane Beryl, second step of the code.



For the wind:

Once founded the minimum, saves the lat/lon values and creates another box around

Source: Made by the author (2025).

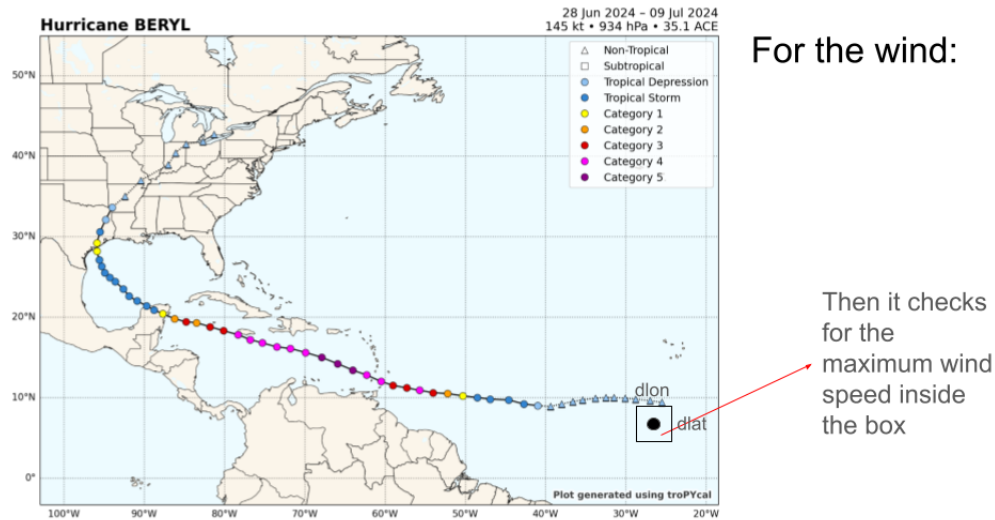
For the second boxes, the following slice of code demonstrates the process.

```
1 # [...] more code before
2
3     upper_lat_WIND = lat_sel + dlat
4     lower_lat_WIND = lat_sel - dlat
5     left_lon_WIND = lon_sel - dlon
6     right_lon_WIND = lon_sel + dlon
```

The first steps are to save the values from lat and longitude tied to the minimum MSLP, and then, the second box is limited by the lines that specifies the WIND parameters, `upper_lat_WIND`; `lower_lat_WIND`; `left_lon_WIND`; `right_lon_WIND`, defined yet again in terms of `dlat` and `dlon`.

Figure B.4 shows the next step, which is searching inside the box for the maximum wind speed.

Figure B.4 - Tracking of Hurricane Beryl, third step of the code.



Source: Made by the author (2025).

The code for this part is the following:

```

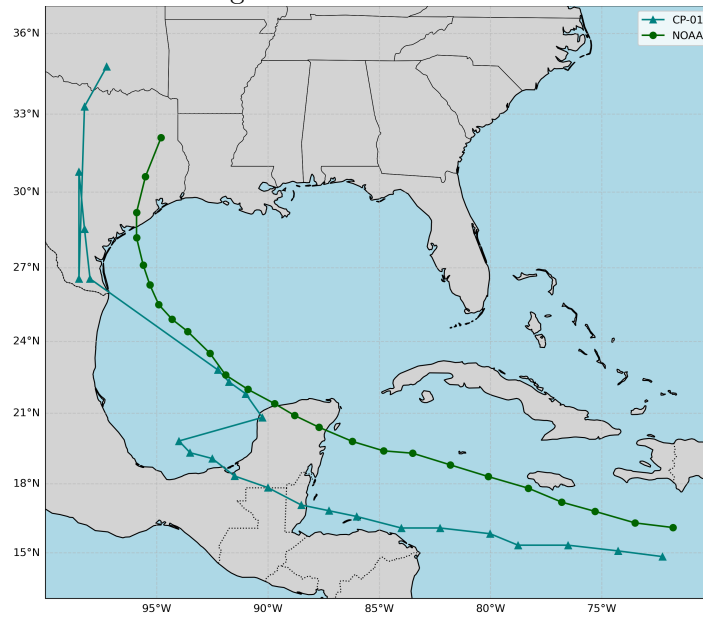
1 # [...] more code before
2
3 # Select WSPD within the new box:
4 if label != 'ERA5':
5     WSPD_sliced = WSPD_t.sel(lat=slice(
6         lower_lat_WIND, upper_lat_WIND), lon=
7         slice(left_lon_WIND, right_lon_WIND))
8 elif label == 'ERA5':
9     WSPD_sliced = WSPD_t.sel(lat=slice(
10         upper_lat_WIND, lower_lat_WIND), lon=
11         slice(left_lon_WIND, right_lon_WIND))
12
13 if WSPD_sliced.size > 0 and not np.isnan(
14     WSPD_sliced).all():
15     # Find the maximum WSPD within the new
16     # box
17     max_wspd = np.nanmax(WSPD_sliced.values)
18     wspd_points.append(max_wspd)
19 else:
20     # If no valid WSPD points, append NaN
21     print(f"No valid WSPD points found for {

```

```
label} at timestep {t}. Skipping...")
wspd_points.append(np.nan)
```

One could ask oneself, what value for `dlat` and `dlon` must be imposed. Some tests were done, and we noticed that we cannot restrict a single value for all types of dataset. For instance, one could look at the trajectory on Figure B.5.

Figure B.5 - Placeholder.

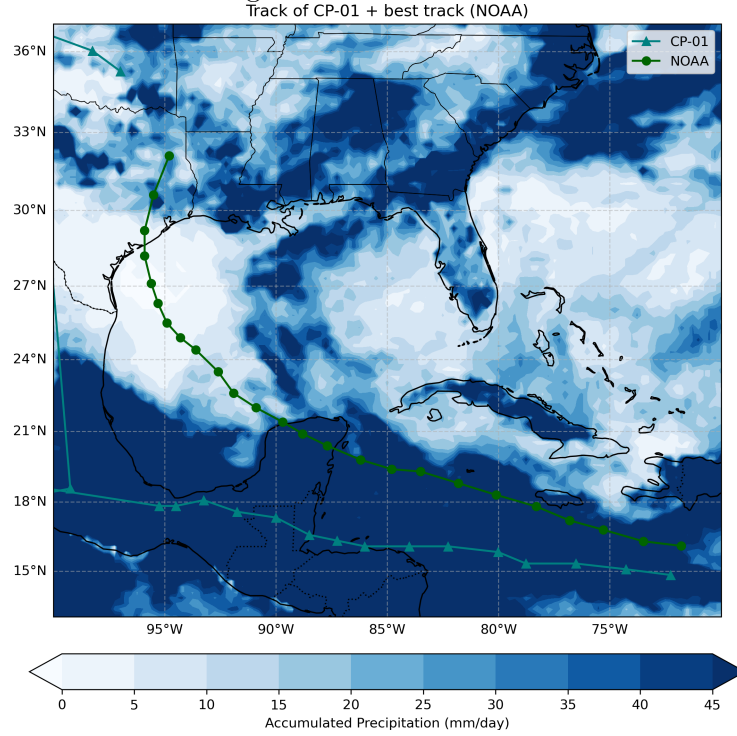


Source: Placeholder.

From what we gather on Figure B.5, the algorithm is able to capture the event, but fails after a certain point.

In order to solve this issue, we plotted the curves generated by the tracking above the rainfall path accumulated on the whole period. Figure B.6 shows this solution:

Figure B.6 - Placeholder.

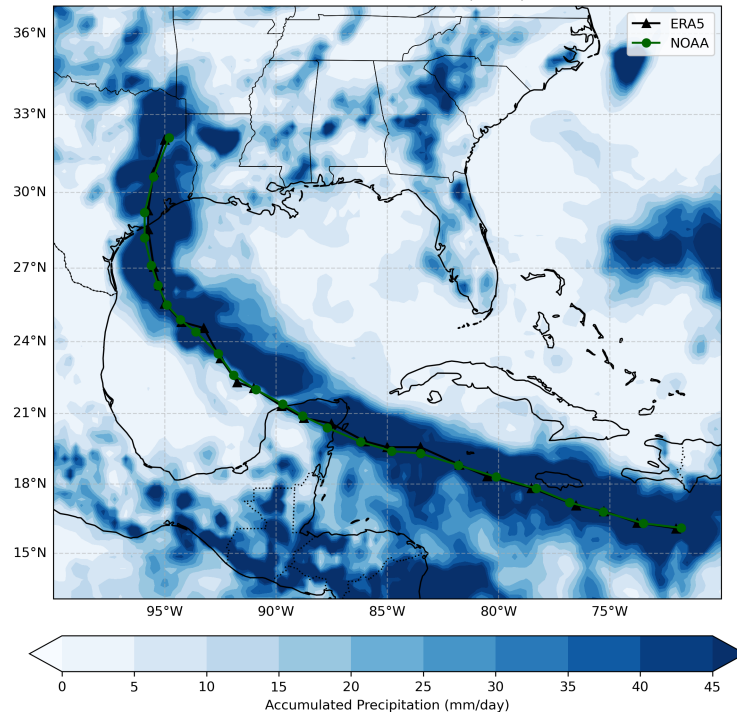


Source: Placeholder.

Based on this correction, we can define the size for `dlat` and `dlon`. We recommend that this process is done, in order to use this tracking.

Furthermore, we can verify, for example, if the behavior of the intensity registered by ERA5 was not related with a possible tracking error. We can confirm this by confirming the trajectory, shown on Figure B.7:

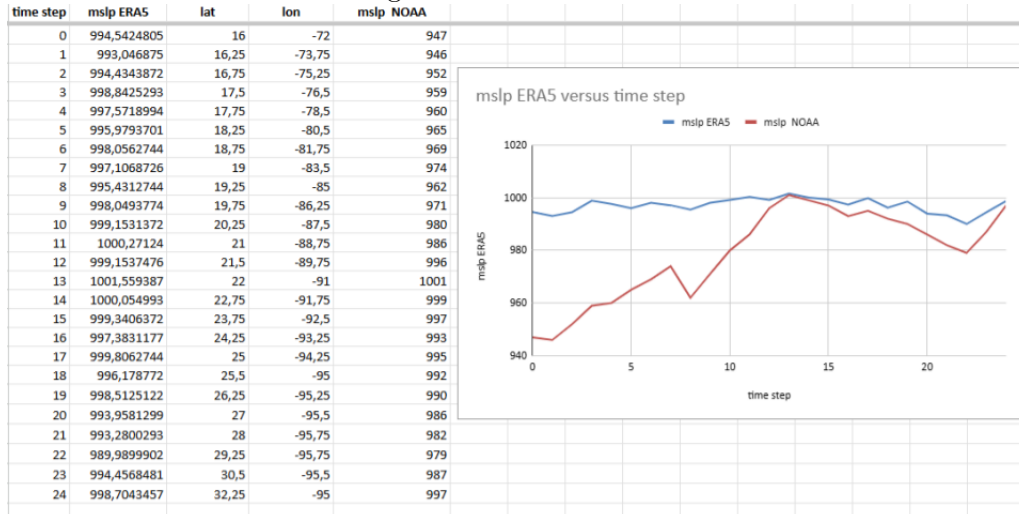
Figure B.7 - Placeholder.
Track of ERA5 + best track (NOAA)



Source: Placeholder.

One could register the associated values to the minimum MSLP and maximum wind speed in a table, and compare it directly with the data from the best track. Figure B.8 shows this table, where the best track data is labeled “NOAA”.

Figure B.8 - Placeholder.



Source: Placeholder.

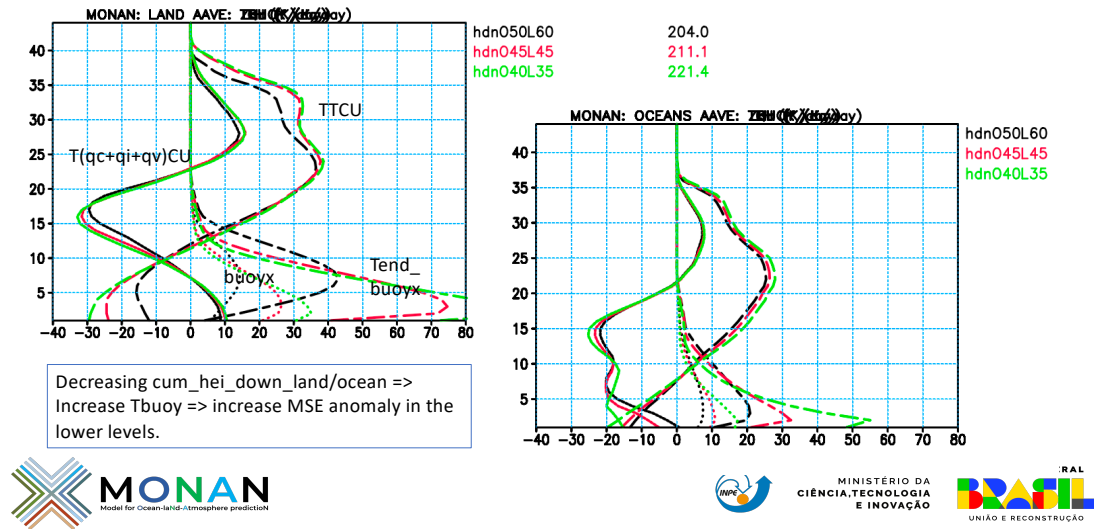
Thus, one can realize that the tracking is in line with the rainfall accumulated trajectory, and the values associated with it are indeed low. Since the tracking searches in a wide box, we believe that the intensity problem from ERA5 is indeed an issue in the reanalysis.

APPENDIX C - DISCUSSION REGARDING THE MAXIMUM DOWNDRAFT HEIGHT

In this appendix, it shall be discussed how the cold pool affects the thermodynamic profile, looking essentially for the temperature and humidity tendency.

Figure C.1 shows the vertical profiles for temperature tendencies (TTCU), humidity tendencies ($T(qc+qi+qv)CU$), buoyancy-excess tendency ($Tend_buoyx$), and the corresponding value of this parameter. The black, pink, and green curves represent distinct configurations by modifying the value for the coefficient, which represents the proximity of the downdraft to the surface, given 0.5 (black curve) being the farthest one from the surface and 0.4 (green curve) being the nearest one. On the left panel, the profile was constructed considering an area of land mass, whereas the right panel was constructed considering ocean mass.

Figure C.1 - Vertical profiles of temperature tendency, humidity tendency, buoyancy-excess tendency and values. On the left panel, over land mass and on the right panel, over the ocean.



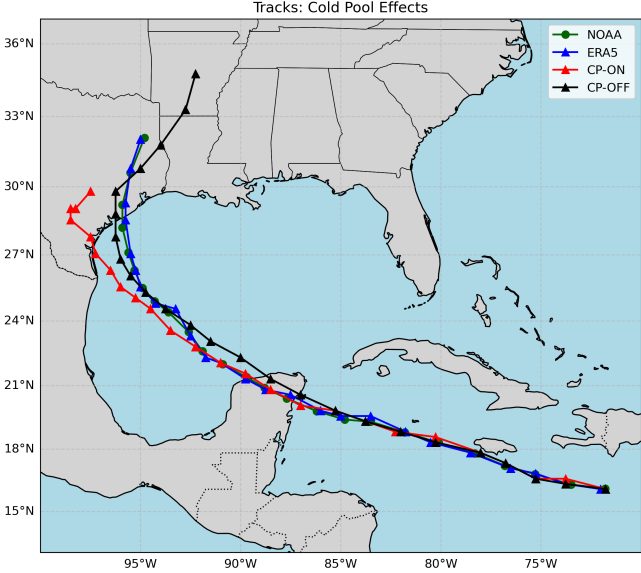
Source: Courtesy of Saulo R. Freitas (2025).

One could notice that the black curve tends to cool down less on lower levels, and warm up less on higher levels. On medium levels, it is possible to notice a less humid layer. The combination of these two tendencies lowers the values of the buoyancy-excess parameter when compared to its peers. In other words, when one puts the downdraft farther from the surface (black curve), the effect of the cold pools is reduced (smaller buoyancy-excess), given lesser cooling on the lower levels and a smaller wetting on the medium levels.

On another note, if one puts the downdraft nearer the surface (green curve), there is a greater cooling on the lower levels, greater humidity on the medium levels, and a warmer layer on the higher levels. This result implies a stronger cold pool, which can be observed both by the tendency of the buoyancy-excess and a greater value for this parameter on lower levels.

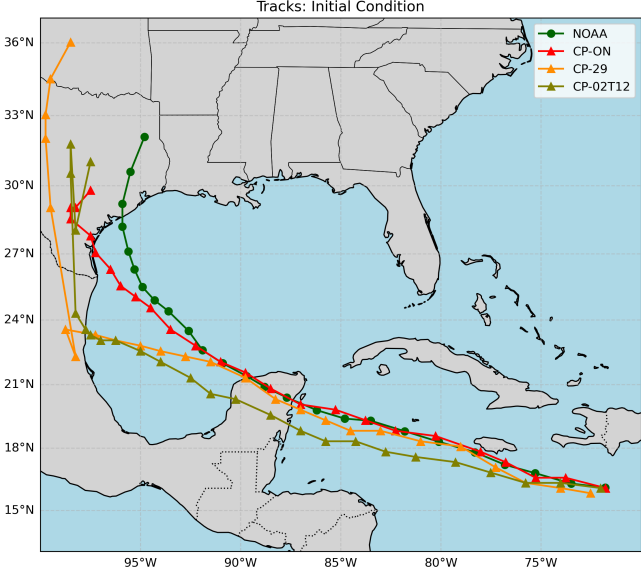
APPENDIX D - SPATIAL PEARSON CORRELATION FOR ALL SIM-
ULATIONS OF BERYL

Figure D.1 - Cold pool effects tracks.



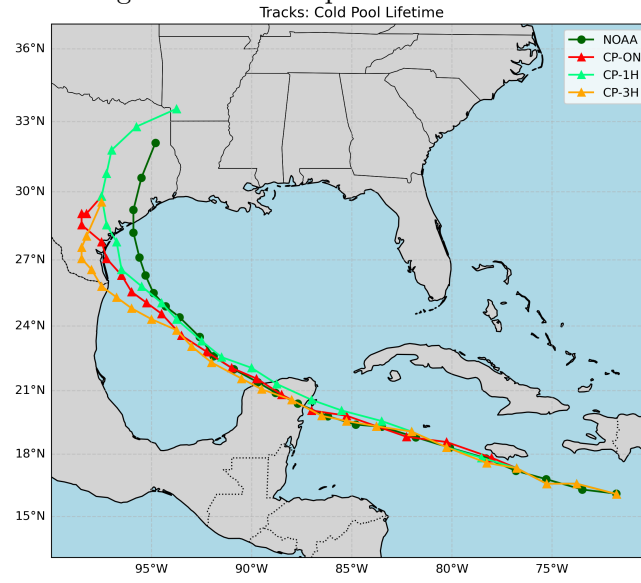
Source: Made by the author (2025).

Figure D.2 - Initial condition tracks.



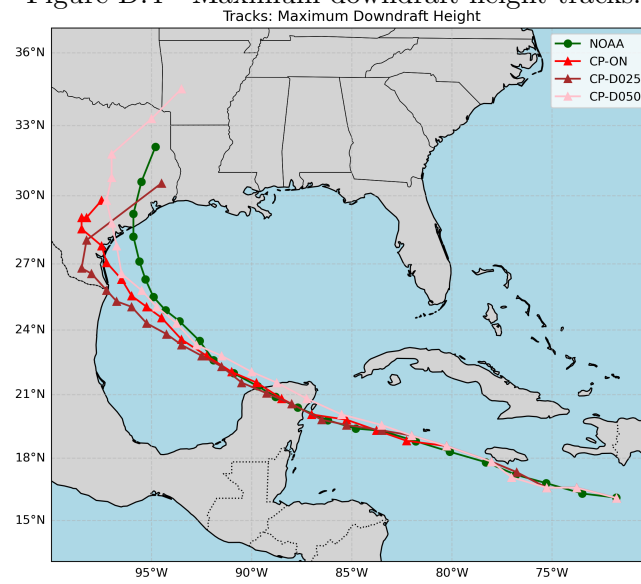
Source: Made by the author (2025).

Figure D.3 - Cold pool lifetime tracks.



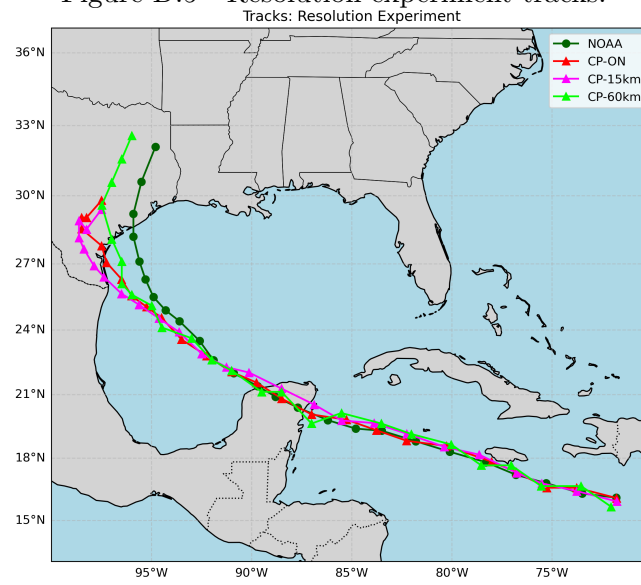
Source: Made by the author (2025).

Figure D.4 - Maximum downdraft height tracks.



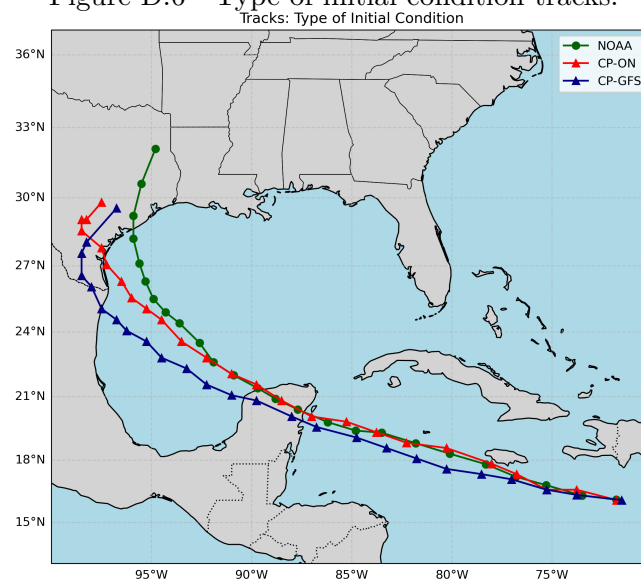
Source: Made by the author (2025).

Figure D.5 - Resolution experiment tracks.



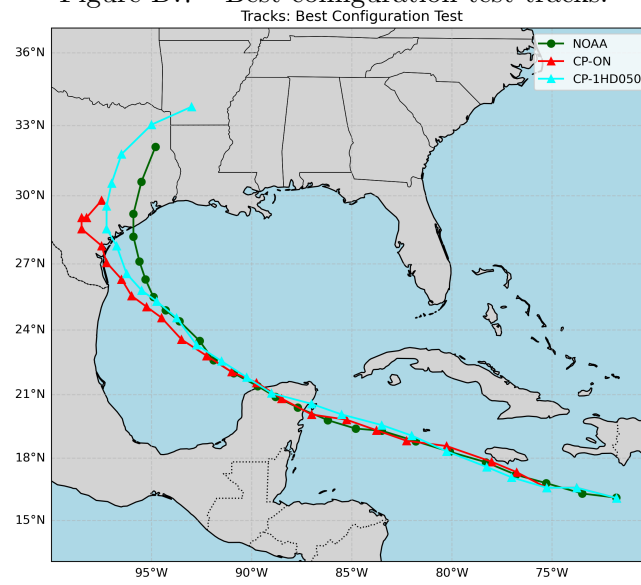
Source: Made by the author (2025).

Figure D.6 - Type of initial condition tracks.



Source: Made by the author (2025).

Figure D.7 - Best configuration test tracks.

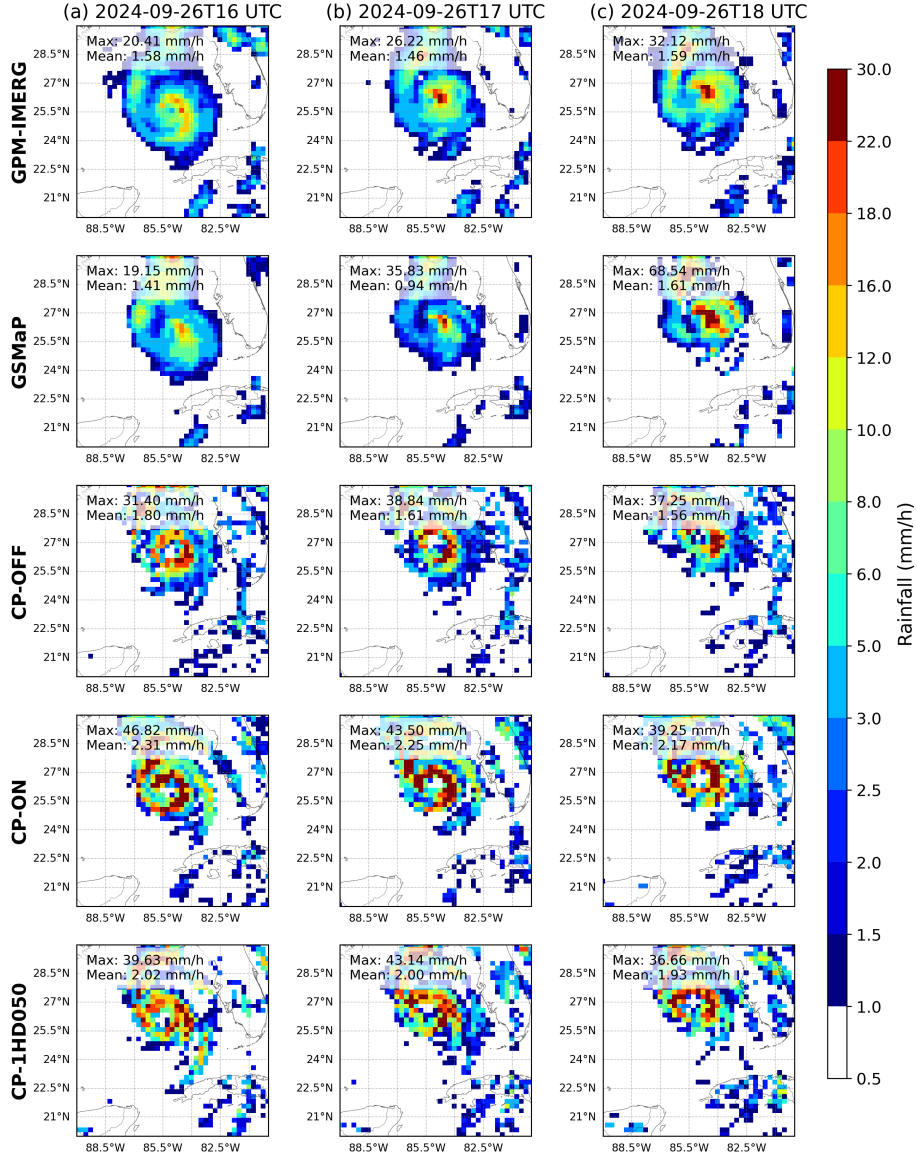


Source: Made by the author (2025).

APPENDIX F - MASK ALGORITHM

Figure F.1 displays the panel computed at the largest ATE error shown in Figure ??.

Figure F.1 - Snapshots of Hurricane Helene at the largest ATE values.

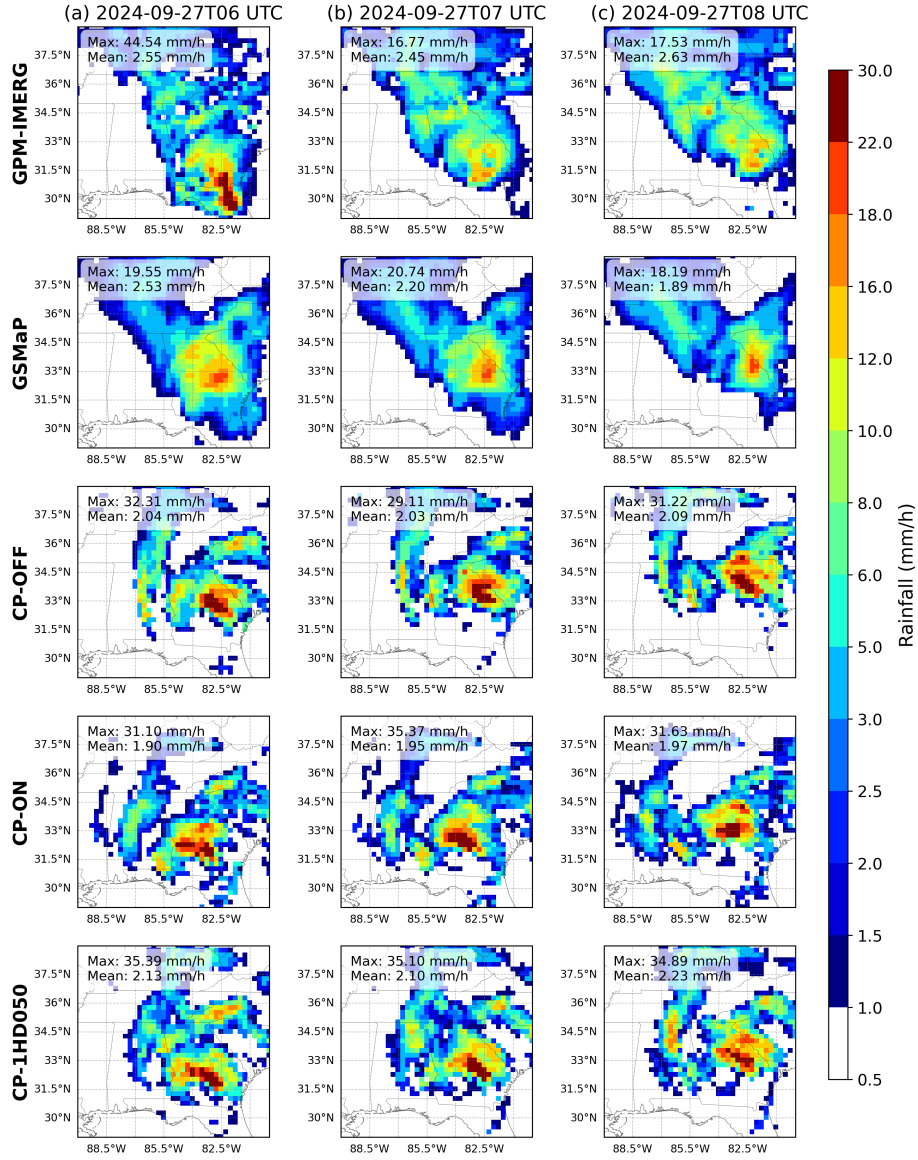


Source: Made by the author (2025).

One can see that in the MONAN simulations, the hurricane has already started to simulate the rainbands that are not showing in the reference datasets, both GPM-IMERG and GSMaP. Besides, there is an error at the rainfall location that is sys-

tematically appearing in this kind of figure, as shown in Figure F.2.

Figure F.2 - Snapshots of the Hurricane Helene phase over land.



Source: Made by the author (2025).

It is possible to observe that the tracking error is very persistent, where the rain location has a slight deviation that can be observed in the panels presented by the MONAN simulation, especially in relation to the GPM-IMERG.

APPENDIX G - MASK ALGORITHM

PUBLICAÇÕES TÉCNICO-CIENTÍFICAS EDITADAS PELO INPE

Teses e Dissertações (TDI)

Teses e Dissertações apresentadas nos Cursos de Pós-Graduação do INPE.

Manuais Técnicos (MAN)

São publicações de caráter técnico que incluem normas, procedimentos, instruções e orientações.

Notas Técnico-Científicas (NTC)

Incluem resultados preliminares de pesquisa, descrição de equipamentos, descrição e ou documentação de programas de computador, descrição de sistemas e experimentos, apresentação de testes, dados, atlas, e documentação de projetos de engenharia.

Relatórios de Pesquisa (RPQ)

Reportam resultados ou progressos de pesquisas tanto de natureza técnica quanto científica, cujo nível seja compatível com o de uma publicação em periódico nacional ou internacional.

Propostas e Relatórios de Projetos (PRP)

São propostas de projetos técnico-científicos e relatórios de acompanhamento de projetos, atividades e convênios.

Publicações Didáticas (PUD)

Incluem apostilas, notas de aula e manuais didáticos.

Publicações Seriadas

São os seriados técnico-científicos: boletins, periódicos, anuários e anais de eventos (simpósios e congressos). Constam destas publicações o Internacional Standard Serial Number (ISSN), que é um código único e definitivo para identificação de títulos de seriados.

Programas de Computador (PDC)

São a seqüência de instruções ou códigos, expressos em uma linguagem de programação compilada ou interpretada, a ser executada por um computador para alcançar um determinado objetivo. Aceitam-se tanto programas fonte quanto os executáveis.

Pré-publicações (PRE)

Todos os artigos publicados em periódicos, anais e como capítulos de livros.